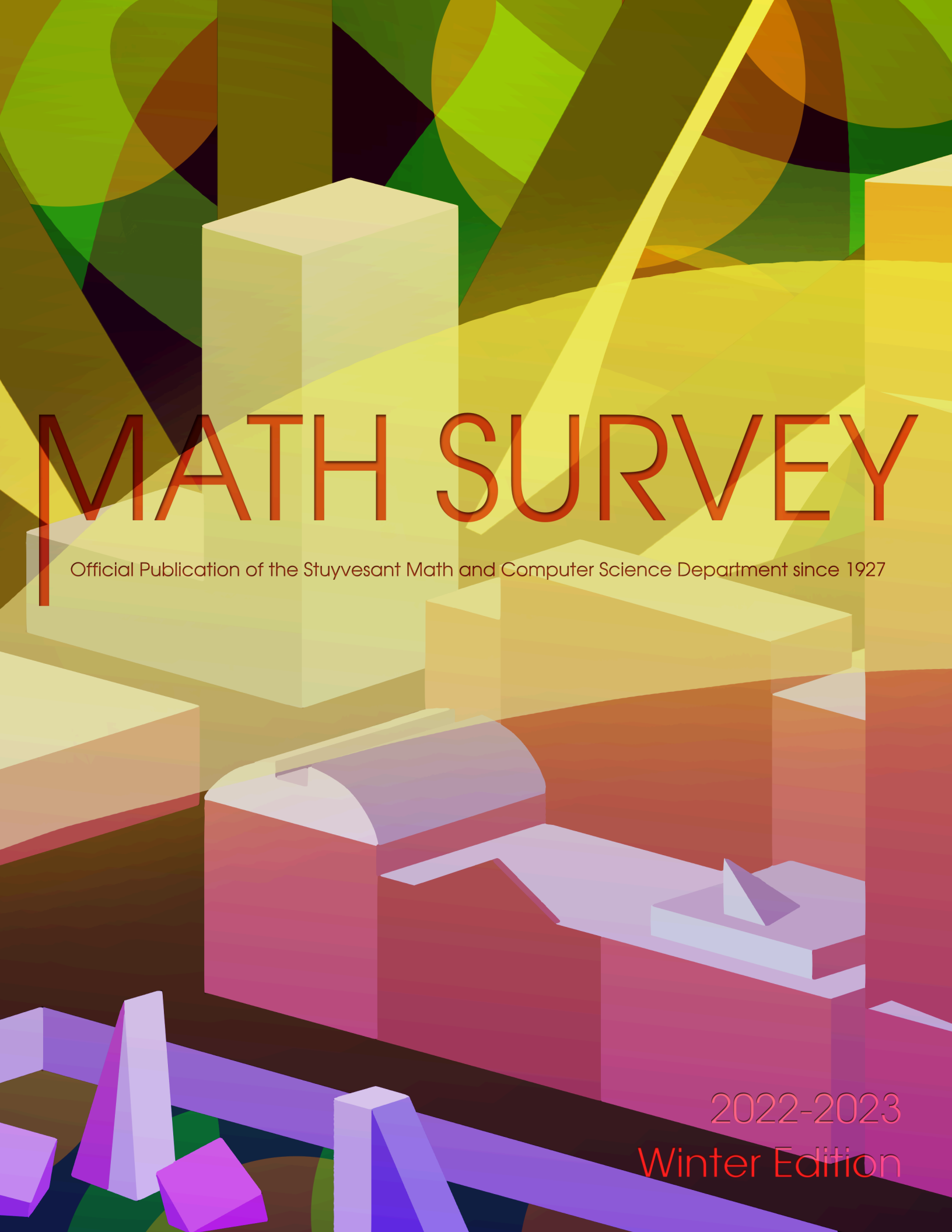




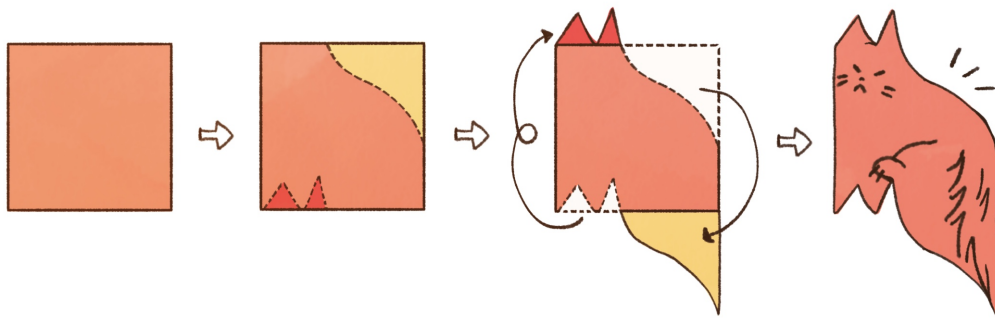
MATH SURVEY

Official Publication of the Stuyvesant Math and Computer Science Department since 1927

2022-2023
Winter Edition

Math Survey

2022-2023 Winter Edition

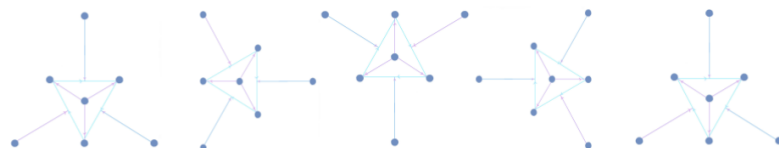


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Math and Computer Science Department since 1927

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Stuyvesant Math Survey

2022-2023 Winter Edition

The Official Mathematics and Computer Science Publication
of the Stuyvesant High School Math Department since 1927

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Special thanks to Professor Joel Lewis (Class of 2003), Mr. JonAlf Dyrland-Weaver (Class of 2001), Professor Jeffrey N. Heinz, Mr. Thomas Lin, and Mr. Konstantin Kakaes for their support for this edition.



Visit <https://stuymathsurvey.com> for the online version.
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Editor's Note

A little bit of history... Stuyvesant High School was established in 1904. In 1927, *Math Survey* was first published, at the time in black and white. Starting in 2001, it was printed in color. After a lapse of a few years, the publication was revived in 2020. It's 2023 now, and we're keeping it going.

The 2022-2023 Winter Edition of *Math Survey* features brainteaser puzzles, articles, interviews, and illustrations. Thank you to our staff members and authors for their time, dedication, and love for mathematics and computer science. I would also like to thank our faculty advisor, Mr. Rubinstein, for his guidance. Finally, we would not be able to print hard copies without the support of the Stuyvesant Parents' Association. Thank you!



RIN FUKUOKA
EDITOR-IN-CHIEF

Foreword

Mathematics, Stuyvesant, and Beyond: Change the Question, Continue Searching for Beauty

Professor Joel Lewis, Class of 2003, has been a mathematics professor at George Washington University since 2017 and has completed notable work in enumerative and algebraic combinatorics. While attending Stuyvesant, he was the captain of the Math Team for two years and also an Intel Science Talent Search finalist (now renamed Regeneron). After graduating from Stuyvesant, he attended Harvard College, and graduated with honors in 2007 as a mathematics major. During his college years, he studied at the Budapest Semester in Mathematics program, and he received Honorable Mentions on The William Lowell Putnam Mathematical Competition in 2004, 2005, and 2006. Lewis earned a Ph.D. from MIT in 2012 with his thesis, “Pattern Avoidance for Alternating Permutations and Reading Words of Tableaux,” under his advisor, Alexander Postnikov. He then worked for five years as a postdoc at the University of Minnesota and continued his illustrious math career at George Washington University.

0

It is incredibly impressive that Stuy students have been writing publishing a serious math magazine for nearly a century—enough that it was intimidating to be invited to write this foreword. I looked at the contributions of other alumni in recent editions for inspiration (they’re at stuymathsurvey.com), but that exacerbated the problem: all those authors are very accomplished, and they’ve already offered a ton of insight and good advice that is hard to match. But hopefully I can write a few reminiscences that are worth reading.

I went to a small middle school and didn’t know how I would fit in at Stuy. It was my good fortune that I loved and was very good at solving math problems, and Stuyvesant is one of the best places in the world to go to high school for someone like that. Don’t get me wrong, there are other good things about Stuy, too—I fondly remember classes with Ms. Thoms and Mr. Badgley—but if I were the kind of person who used the phrase “spiritual home,” I would say that my spiritual home is the 4th floor at 345 Chambers Street, and especially the Stuyvesant Math Team.

0.5

An ant moves along the edges of a tetrahedron. Whenever it reaches a vertex, it chooses one of the three outgoing edges at random. What is the probability that it is back where it started after 17 steps?

What if the ant never walks along the same edge twice in a row?

1

I started as Math Team captain on Tuesday, September 4, 2001. The first thing I remember about that semester is that my schedule had English in the slot when math team met, and Mr. Geller had to go to the program office to get it changed for me. The second thing I remember is that there were more than 100 freshmen on the math team; most of them were in the auditorium, while I had a group of 30 or so in a classroom. The third thing I remember about that semester is being in my (now rescheduled) English class on Tuesday, the second week of the semester; hearing a loud noise from outside; and then later a student running in and telling us that a plane had flown into the World Trade Center.

There's too much about that one day for this short note; suffice it to say that the world changed a lot in the course of a few hours. My GW undergraduates are a few years older than most of you, so I've just recently passed the point where the seminal moments of my life happened before my students were born. Maybe one of you will find yourself writing about going to Stuy via Zoom for the 2042–2043 edition of *Math Survey*—good luck!

We were out of school for—a few days? a week? Then we were relocated to Brooklyn Tech. The Tech students started school in the early morning and finished around noon, then we came in and stayed until the early evening. Periods were a shorter than usual, which was stressful for everyone. In early October, we returned to Stuy. On Chambers Street, we could smell the fires that were still burning underground.

1.5

Suppose that a point (x, y) is to be chosen so that it lies inside or on the boundary of the ellipse $x^2 + 3y^2 = 17$. What is the largest possible value of $3x^2 + y^2$? The smallest?

2

My first experience with “real math” came via the Math Research class. I started working with Joel Spencer, a professor at NYU, during my junior year. Joel (what I call him now—of course back then he was Professor Spencer) gave me some problems to solve (Putnam exam questions and the like), and then I came back the next week and told him my solutions, and then he gave me some more problems to solve, and taught me a little bit about binary search and Ulam's game (see [1]). After a little while of this, he gave me a problem to solve that no one had ever solved before. And that's all math research is, really.

Of course the difference between solving problems designed to be solved in 10 minutes and solving problems whose solutions aren't known is that the second kind of questions are often a lot harder. (Not always: sometimes problems are unsolved just because no one ever asked them before. But often.) So instead of getting a rush of success after a few minutes of struggle, you can spend an hour ... or a day ... or a week ... or a month not making much progress at all.

I remember filling up pages of notebooks with calculations related to Joel’s problem, just trying to find a nice pattern. It was hard! In fact, I never solved the problem he gave me—the patterns weren’t nice enough for the tools I had available. (It was solved later by Joel and Ioana Dumitriu [2].) Instead, I noticed that if I changed the rules of the game slightly, suddenly there were nice numbers floating around, formulas that could be guessed and proved by induction. Joel told me to forget about the original question and work on the new one.

There may be good life lessons in this story. (Perhaps, not to be afraid of initial failures, and to seek out advice and mentorship.) But for me it is mostly about something I love about abstract mathematics: when you find yourself at a dead end, working with objects that are difficult or ugly, you can always *change the question* and go look for something beautiful instead.

2.5

There is a line that is tangent to the graph $y = x^4 + 2x^3 - 3x^2 + 2x + 1$ at two different points. What are these points?

Which quartic curves have such a “double tangent line”?

3

Let me end with one concrete piece of advice. Many of you will find yourself in college at a place that is even larger and less personal than Stuyvesant. Your professors will have two or three hours a week called office hours. If you are struggling with a class, *go to office hours!* Talk to the professor about the course material, what’s giving you trouble, and what to do differently. But, let’s be honest: you’re Stuy students, so you might not find yourself struggling too often.¹ Suppose you’re doing well in your classes; in that case, *go to office hours anyhow!* Tell the professor you’re interested in the material and ask for one or two suggestions of something extra to read. Don’t be embarrassed or shy; the best part of my job is talking about mathematics with engaged students, and most of my colleagues feel the same way. (The other ones won’t remember who you are, so there’s not harm in trying.)

3.5

A *swap* in a list takes two adjacent elements and interchanges them, leaving all the others untouched. What is the fewest number of swaps needed to sort the list $(5, 1, 3, 6, 2, 4)$ into its natural order $(1, 2, 3, 4, 5, 6)$?

Suppose I loosen the notion of swap, so you can swap any two entries in the list in one step. (For example, now there are $\binom{6}{2} = 6C2 = 15$ different swaps that can be performed on $(5, 1, 3, 6, 2, 4)$, and one of them produces $(2, 1, 3, 6, 5, 4)$.) How many swaps do I need now?

¹Let’s put the fact that my college freshman-year transcript includes a B–, a C+, and an F somewhere innocuous, like in a footnote. (Needless to say, I did not make good use of office hours in those classes!)

Now suppose the cost of a swap is the distance between the swapped entries (so that swapping the 2 and 5 in $(5, 1, 3, 6, 2, 4)$ has cost 4, because they are 4 spaces apart)—what’s the smallest total cost to put it in order?

What if you start with other permutations?

4

Some comments on the questions:

To solve the first question, the first step is to make it harder: set $17 = n$ and try to solve the same question in terms of n . You can probably think of other nice ways to generalize it. If you like this question, you might be interested in learning more about *Markov chains* or the *transfer-matrix method*.

The second and third questions are deceptive. Students in Ms. Pascu’s multi class should know one set of techniques for approach the second question—but that’s not necessarily the most elegant way to do it. Good luck to solving the third question (which I modified from ARML 2003) with calculus; try to think of an elementary idea.

The last question is near and dear to my heart. The first version is a classic piece of combinatorics that was understood in the 19th century. It is also related to the *bubble sort* algorithm from computer science; the keyword to search for is *inversions*. For the second version, it is helpful to think of a permutation as a function (so $(5, 1, 3, 6, 2, 4)$ represents the function f such that $f(1) = 5, f(2) = 1, \dots$) and to consider its *cycles* (like $1 \xrightarrow{f} 5 \xrightarrow{f} 2 \xrightarrow{f} 1$) and to work out what happens to the cycles when you perform a single swap. This was studied by Dénes in the 1950s [3] but may have been known earlier. The third version, which is related to the *straight selection sorting algorithm*, is much newer; it was introduced and solved by Petersen and Tenner in 2015 [4].

Joel Lewis
January 2023

References

- [1] *Ulam’s game* on Wikipedia. https://en.wikipedia.org/wiki/Ulam%27s_game
- [2] Dumitriu, Ioana; Spencer, Joel. A Halfliar’s game. *Algorithmic combinatorial game theory. Theoret. Comput. Sci.* 313 (2004), no. 3, 353–369.
- [3] Dénes, József. The representation of a permutation as the product of a minimal number of transpositions, and its connection with the theory of graphs. *Magyar Tud. Akad. Mat. Kutató Int. Közl.* 4 (1959), 63–71.
- [4] Petersen, T. Kyle; Tenner, Bridget Eileen. The depth of a permutation. *J. Comb.* 6 (2015), no. 1-2, 145–178.

Tracing Pictograms

David Jiang

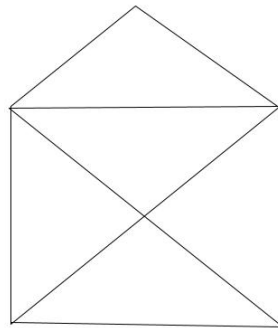


Am I an Artist?

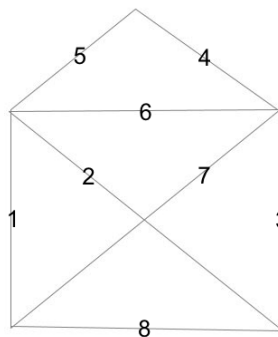
The following puzzles require you to trace the figure

- Without lifting your pen/pencil
- Without going back over what you've already drawn.

Take, for example, the following pictogram:

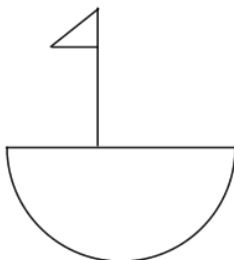


One can use this sequence to trace over the drawing:

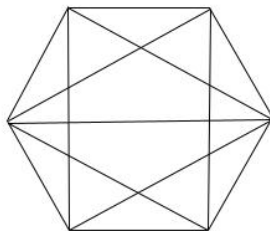


For the pictograms below, they can either be traced using the above rules or they are impossible to draw. If they're impossible, show why a solution does not exist.

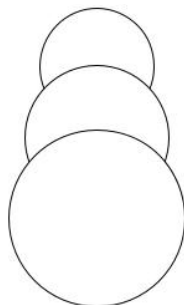
Pictogram #1



Pictogram #2



Pictogram #3



The solution is on page 9.

Alice in Subtractionland

Aditya Pahuja



Alice and Bob are playing the so-called “subtraction game.” In this game, there are n tokens in a single pile, and each player, starting with Alice, takes turns removing between 1 and k tokens from the top of the pile. The goal is to be the last person to pick up a token.

(a) Given any n and k , determine which player wins with optimal play, and describe a simple algorithm by which to make the optimal move every time.

(b) Suppose there are instead two piles, perhaps with differing numbers of tokens, and Alice and Bob may only remove tokens from one pile during their turn. Who wins then?

(c) Now, suppose each token has a weight $w \leq k$ less than k . The players are allowed to remove some subset of tokens from the top of the pile whose weights have a sum that is at most k . Describe an algorithm for optimal play. Is the outcome of the game still determined solely by the choice of n and k ?

The subtraction game is one of many variants of the game Nim (confusingly, the subtraction game itself is also sometimes referred to as “Nim”). Nim itself involves multiple piles of tokens, and the players are allowed to remove any number of tokens from each pile, with the same goal of being the last player to make a move.

Interestingly, it can be shown that the subtraction game and the presented variants are equivalent to a game of Nim, for appropriately chosen values of p and $n_1, n_2, \dots, n_p$; this is a result of the [Sprague-Grundy theorem](#).

The solution is on page 11.

The 2023 Puzzle

Amanda Louie



1 The 2023 Puzzle

Use all of the digits in 2023 (2,0,2,3) to create expressions that result in the numbers from 1-30.

1.1 Rules

Use arithmetic operations (addition, subtraction, multiplication, division), factorial, exponentiation, and/or concatenation (e.g., $52 = 22 + 30$, but not $(\times * 2 + 0)3 = 43$) to find as many solutions as possible. Use each digit only once.

Example: To produce 1, you can do $(2 \times 0) + (3 - 2) = 1$

1 =
2 =
3 =
4 =
5 =
6 =
7 =
8 =
9 =
10 =
11 =
12 =
13 =
14 =
15 =
16 =
17 =
18 =
19 =
20 =

21 =
22 =
23 =
24 =
25 =
26 =
27 =
28 =
29 =
30 =

The solution is on page 13.

Word Scramble!

Anastasia Lee



How well do you know your math terms? Find out by completing this puzzle! For each line, unscramble the letters to form a well-known math-related word/phrase. Then use the hint below it to locate the letter at that index (0-indexed!). Unscramble these letters to get the final word! Apostrophes are implied where necessary.

1. aonlioswsfce

The second smallest number such that $\phi(n) = 6$, i.e., there are 6 integers less than n that are relatively prime to n .

2. slcaidiraxa

When written in base 3, it is the smallest integer that has six divisors.

3. iosceyyhdikkettinct

A number that is the product of three numbers and the sum of those same three numbers.

4. eaosesmwhtrttrter

The product of 15 consecutive numbers that sum to 90.

5. ncvtroiute

Positive root of $x^3 - 2x^2 - 7x - 4$.

6. lriendyutesiet

A number that is equal to the sum of 3 raised to its digits.

7. inrhalyqayts cuzihcuwea

The difference between the first two perfect numbers (positive integers equal to the sum of their proper divisors).

8. rcceimcliucr

$11x$, where x is the expected number of (fair) coin flips needed to get heads.

9. litgreaypahoptner

How many times more subsets can be formed from a set of size 26 than a set of size 23.

10. nltacirgeebso

The number for which the sum and difference of its two prime divisors are both prime.

The solution is on page 14.

Estimathon!

Anastasia Lee



How to Play

This puzzle is a test of your estimation skills! For each of the following questions, construct an interval that you think will contain the correct answer. (If it does, it is considered **good**.) When you finish, calculate your score as:

$$\left(10 + \sum_{\text{good intervals}} \left\lfloor \frac{\max}{\min} \right\rfloor \right) \cdot 2^{10 - \# \text{ of good intervals}}.$$

The goal is to have the lowest score. Challenge a friend and see who wins!

Questions

1. How many Earths would fit inside the Sun?
2. How many smartphones were sold worldwide in 2022?
3. How many Starbucks locations are there in New York City?
4. How long is the Hudson River (in feet)?
5. How expensive was the new Stuyvesant High School campus (when it was built)?
6. How many intersections in New York City have a traffic signal?
7. Across all its locations, how many items are housed in the New York Public Library?
8. How many acres of New York City are parks?
9. How many gold bars are stored in New York City's Federal Reserve Bank?
10. How many pounds does the Statue of Liberty weigh?

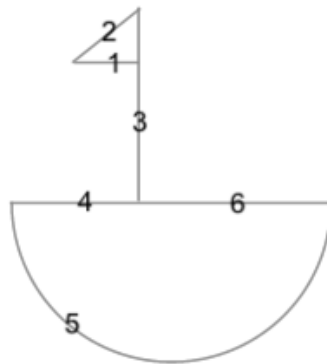
The solution is on page 15.

Solution to Tracing Pictograms

David Jiang

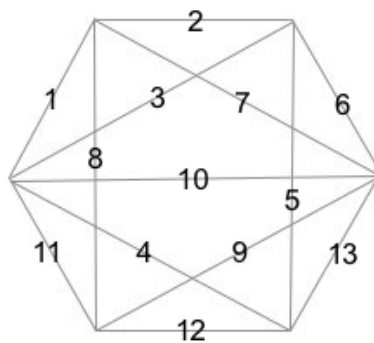
Pictogram #1

You can trace the figure in the following way:



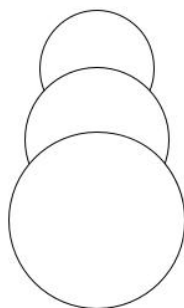
Pictogram #2

You can trace the figure in the following way:



Pictogram #3

The figure is impossible to trace.



Notice that if we convert this into a graph, we get four vertices, each with degree three. When tracing over this figure, we must always enter a vertex from one edge and exit from another. Therefore, each vertex that is not the starting or ending point must have even degree.

Thus, there can be at most two vertices with odd degree. However, in this graph, there are four vertices with odd degree, so we cannot trace the figure. How can the figure be changed so that a solution is possible?

Solution to Alice in Subtractionland

Aditya Pahuja

(a) Alice wins whenever $k + 1$ does not divide n , and Bob wins otherwise.

Our first observation is that this game is entirely symmetric: the outcome depends only on whom the first player is. Therefore, we can let a $\mathcal{P}$ -position be a game state for which the current player wins, while an $\mathcal{N}$ -position is a game state for which the next player to move wins. A crucial consequence of this is that if the game state is currently a $\mathcal{P}$ -position, then it will be an $\mathcal{N}$ -position on the next move, and vice versa.

Now, let S denote the starting state of the game, and consider the case where $n \leq k + 1$ as an example.

1. When $n < k + 1$, Alice can simply remove the entire pile to win. This means S is a $\mathcal{P}$ -position.
2. When $n = k + 1$, Alice's move removes up to k tokens, so there will be between 1 and k remaining tokens. We saw above that this is a $\mathcal{P}$ -position, which means the previous game state, S , was an $\mathcal{N}$ -position; hence, Bob wins.

We can inductively apply this logic to generalize. Suppose $n = q(k + 1) + r$ for integers q, r with $0 \leq r < q$. Then, we induct on q . The base case where $q = 0$ was dealt with above, so we consider some arbitrary choice of q , with the claim that S is a $\mathcal{P}$ -position if and only if $r \neq 0$. If this is indeed true for this choice of q , then consider what happens when q is incremented — that is, $k + 1$ new tokens are added, so $n = (q + 1)(k + 1) + r$. We analyze $r = 0$ and $r \neq 0$ separately now:

1. If $r = 0$, Alice's move leaves $n' = q(k + 1) + r'$ tokens behind, and r' cannot be 0 since this would require a removal of $k + 1 > k$ tokens. Hence, Alice's move results in a $\mathcal{P}$ -position, meaning the original position is an $\mathcal{N}$ -position.
2. If $r > 0$, then Alice can remove r tokens. Then, this is the game state where $r = 0$, which is an $\mathcal{N}$ -position, so S is a $\mathcal{P}$ -position.

Thus, the truth of the claim for q implies the truth of the claim for $q + 1$, so the principle of mathematical induction ensures that the initially claimed answer is indeed correct.

(b) First, we resolve the case where we have two piles of equal length. Here, Bob can adopt the simple strategy of “copy Alice's move,” which always wins. Thus two piles of equal length is always an $\mathcal{N}$ -position.

Now suppose one pile has a tokens while the other has $a + b$ tokens. This game then becomes equivalent to the one-pile case with $n = b$, with the caveat that a player can try “wasting a move” by removing tokens from the pile with a tokens. However, this is only favorable when the game is in an $\mathcal{N}$ state, at which point the other player can also “pass a move” by removing the same

number of tokens from the other pile, reverting back to the same scenario.

Therefore, a player wins the two-pile variant if and only if they win the one-pile variant with b tokens.

(c) We describe a recursive algorithm to make the optimal move. Let the m bottom-most weights be $w_1, w_2, \dots, w_m$, with m being the smallest possible number such that

$$w_1 + \dots + w_m > k.$$

A player is winning if and only if they can force these m tokens to be the only remaining tokens during the other player's turn. Then, we see that winning the game is equivalent to winning the game without those m tokens, so follow this same procedure to make the optimal move after removing those tokens.

For the base case, when the total weight of the tokens is a positive number at most k , remove all the tokens. If the total weight is zero, remove any number of tokens. (In fact, the total weight being zero after this reduction is a lose condition, so it doesn't matter what you do.)

Also, the game's outcome is not uniquely determined by n and k ; one such instance with variable results is $n = 6$ and $k = 2$. When the tokens are 1, 2, 2, 1 in order, Bob wins, but when the tokens are 1, 1, 2, 2 in order, Alice wins.

A Few Solutions to the 2023 Puzzle

Amanda Louie

There are multiple solutions to this puzzle. Here are some:

$$1 = (2 \times 0) + (3 - 2)$$

$$2 = 2 \times (3 - 2 - 0)$$

$$3 = 3 + (0 \times 2 \times 2)$$

$$4 = 3 + (2 \div 2) + 0$$

$$5 = 3 + 2 + (2 \times 0)$$

$$6 = 3 \times 2 + (2 \times 0)$$

$$7 = 3 + 2 + 2 + 0$$

$$8 = 3 * 2 + 2 + 0$$

$$9 = 3 \times [(2 \times 2) - 0!]$$

$$10 = 3^2 + 2 - 0!$$

$$11 = 3^2 + 2 + 0$$

$$12 = 3^2 + 2 + 0!$$

$$13 = 3 \times (2 + 2) + 0!$$

$$14 = 2^{(3+0!)} - 2$$

$$15 = 3 \times (2 + 2 + 0!)$$

$$16 = 2 \times 2 \times (3 + 0!)$$

$$17 = 2^3 \times 2 + 0!$$

$$18 = (3 + 0!)^2 + 2$$

$$19 = 22 - 3 + 0$$

$$20 = 20 \times (3 - 2)$$

$$21 = 20 + 3 - 2$$

$$22 = 22 + 3 \times 0$$

$$23 = 23 + 2 \times 0$$

$$24 = 22 + (3 - 0!)$$

$$25 = 22 + 3 + 0$$

$$26 = 22 + 3 + 0!$$

$$27 = (2 + 2)! + 3 + 0$$

$$28 = (3 + 0!)! + 2 + 2$$

$$29 = 30 - (2 \div 2)$$

$$30 = 30 \times (2 \div 2)$$

Solution to Word Scramble!

Anastasia Lee

1. After unscrambling: Law of Cosines

We can see that 7 is the smallest number such that $\phi(n) = 6$, and then $\phi(8) = 4$, but $\phi(9) = 6$. The 9th letter is N.

2. After unscrambling: Radical Axis

The smallest integer that has six divisors is 12, and $12_3 = 5$. The 5th letter is A.

3. After unscrambling: Hockey Stick Identity

The only number that is the product of three numbers and the sum of those same three numbers is $6 = 1 \cdot 2 \cdot 3 = 1 + 2 + 3$. The 6th letter is S.

4. After unscrambling: Stewart's Theorem

The average of the 15 consecutive numbers is $\frac{90}{15} = 6$, and this must be the middle number, meaning that the smallest number is -1 . But then 0 is one of the numbers, so the entire product is 0. The 0th letter is S.

5. After unscrambling: Unit Vector

We can see that -1 is a root, as $-1 - 2 + 7 - 4 = 0$, so $x + 1$ is a factor. Using polynomial long division or synthetic division, we get $x^3 - 2x^2 - 7x - 4 = (x + 1)(x^2 - 3x - 4)$. This can simply be factored as $(x + 1)(x + 1)(x - 4)$, so $x = 4$ is the positive root. The 4th letter is V.

6. After unscrambling: Euler's Identity

We can see that none of the one-digit numbers will work, since $n < 3^n$ for all n . Moving on to two-digit numbers, $10 > 3^1 + 3^0$ and $11 > 3^1 + 3^1$, but $12 = 3^1 + 3^2$. The 12th letter is T.

7. After unscrambling: Cauchy-Schwarz Inequality

The first perfect number is 6 ($1 + 2 + 3 = 6$), and the second perfect number is 28 ($1 + 2 + 4 + 7 + 14 = 28$). The 22nd letter is Y.

8. After unscrambling: Circumcircle

We can express x recursively. If the first flip is heads, then we are done. If not, the expected number of flips to get heads is still x , but having used one flip already, the total will be $x + 1$. So, we get $x = \frac{1}{2} \cdot 1 + \frac{1}{2}(x + 1)$. Solving gives $x = 2$, so $2x = 4$. The 4th letter is U.

9. After unscrambling: Pythagorean Triple

For a set of size n , there are 2^n possible subsets, since you can choose whether or not to include each element. So, the index is $2^{26}/2^{23} = 2^3 = 8$. The 8th letter is E.

10. After unscrambling: Angle Bisector

The smallest potential answer is $2 \cdot 3 = 6$, but $3 - 2 = 1$, which is neither prime nor composite. The next smallest is $2 \cdot 5 = 10$, which works because $5 - 2 = 3$ and $5 + 2 = 7$ are both prime. The 10th letter is T.

Unscrambling the letters N, A, S, S, V, T, Y, U, E, T, we get STUYVESANT as our final answer.

Solution to Estimathon!

Anastasia Lee

1. 1.3 million
2. 1.21 billion
3. 689 locations
4. 1,663,200 feet
5. \$150 million
6. 13,543 intersections
7. 56 million items
8. 28,000 acres
9. 497,000 gold bars
10. 450,000 pounds

Numbers on a Clothesline

Aditya Pahuja



“A generating function is a clothesline on which we hang up a sequence of numbers for display.”
Herbert Wilf, *generatingfunctionology*

1 Introduction

In mathematics, one often deals with structures that can be mapped to the natural numbers; these may be in the form of sequences or sets of things, to give some examples. Generating functions are a powerful way to work with such structures, providing an algebraic lens with which to view combinatorial or number-theoretic objects. Indeed, we will see that this is where their power lies, as the ability to swap between algebraic and combinatorial thinking provides insights that are tremendously difficult to reach otherwise.

2 A Look at the Binomial Theorem

Theorem. The expression $(x + y)^n$ can be written as

$$\sum_{k=1}^n \binom{n}{k} x^k y^{n-k}.$$

Proof. For each factor in the expression $(x + y)^n$, we can choose whether the product contributes an x or a y ; making this choice for every factor generates some term of the form $x^j y^{n-j}$, depending on how many x s and y s were chosen.

Now, consider the coefficient of the $x^k y^{n-k}$ term. We want to choose k of the $x + y$ factors to take an x from, and fix the other $n - k$ factors to contribute a y , since this is the only way to get an $x^k y^{n-k}$ term. There are $\binom{n}{k}$ ways to do this, which means the product generates $\binom{n}{k}$ terms of the form $x^k y^{n-k}$, i.e., the coefficient of $x^k y^{n-k}$ is $\binom{n}{k}$, as desired.

The binomial theorem by itself is already a very useful theorem, and one that will feature prominently later. However, there's something more to be said about the concept shown here. Specifically, let's slightly modify the product to look like

$$\prod_{i=1}^n x_i + 1 = (x_1 + 1)(x_2 + 1) \cdots (x_n + 1).$$

What we have done here is set $y = 1$ and then given a label to each of the factors. Before we do anything, the first question we should think about is, what does the product look like when expanded? (Try it yourself!)

In the case of $n = 3$,

$$(x_1 + 1)(x_2 + 1)(x_3 + 1) = x_1x_2x_3 + x_1x_2 + x_2x_3 + x_1x_3 + x_1 + x_2 + x_3 + 1.$$

The interesting observation here is that each subset of $\{x_1, x_2, x_3\}$ appears within this sum exactly once: $\{x_1, x_2\}$ appears as x_1x_2 , $\{x_2\}$ as x_2 , and so on.

Now, how is this related to the binomial theorem? Let's take all the terms that have k x s, ignoring the numbers assigned to the individual x s. How many of these are there? Well, from the same argument, there are $\binom{n}{k}$ of these terms (which makes sense, considering we're counting subsets). This means that, in a sense, $(x + y)^n$ was counting precisely the same thing: the number of subsets of an n -element set.

The crucial idea captured here is that the coefficients of $(x + y)^n$ are **filtering the subsets** of $\{1, 2, \dots, n\}$ by the number of elements they have. This is the guiding principle behind generating functions as a whole: *Generating functions are algebraic objects that filter information.*

3 Counting with Functions, Part 1

This is the “classical example” of a combinatorial problem that's really hard to solve without a generating function.

Problem. Let $n = pk$ for an odd prime p and positive integer k . Given a set $S = \{1, 2, \dots, n\}$, find the number of subsets of S which have a sum of elements divisible by p .

Solution. First, what makes this a hard problem? The key difficulty is that normally, when dealing with the sum of a set's elements, one looks at more *global* aspects of the setup, such as the sum of all elements within subsets of a fixed size. Controlling the sum of a particular subset's elements is tricky with standard combinatorial methods.

This is where generating functions come in. Rather than filtering by the number of elements, we instead want to filter by the sum of the elements. This motivates the function

$$F(x) := (x + 1)(x^2 + 1)(x^3 + 1) \cdots (x^n + 1).$$

When expanded, the factor $x^j + 1$ either contributes 0 or j to each term, so each the degree of each term corresponds to the sum of the elements of the subset that generated this term. As before, the degree of each term is what allows us to filter for certain information. Now, we need some way to

get at the terms with degrees that are multiples of p .

The key insight to finish here is that, if we take $\zeta \neq 1$ to be a complex number with $\zeta^p = 1$, then

$$1 + \zeta + \zeta^2 + \cdots + \zeta^{p-1} = 0,$$

since $\zeta^p - 1 = (\zeta - 1)(1 + \zeta + \zeta^2 + \cdots + \zeta^{p-1}) = 0$. Using this, one can determine that

$$\frac{F(1) + F(\zeta) + F(\zeta^2) + \cdots + F(\zeta^{p-1})}{p}$$

is the answer, which turns out to be

$$\frac{2^{pk} + (p-1)2^k}{p},$$

with the computation details left as an exercise to the interested reader.

3.1 Practice Problems

1. A fair 6-sided die is rolled thrice. What is the probability that the sum of the three rolled numbers is 13?
2. A 6-sided die is rigged so that the probability of rolling k is $\frac{k}{21}$. If the die is rolled thrice, what is the probability that the sum of the rolled numbers is 13?

4 The Algebraic View

As we've seen, generating functions can be very powerful tools for filtering information about combinatorial setups where the objects can be partitioned into groups corresponding to integers. So far, however, our use of these objects has been limited to scenarios with finitely many objects. The theory we will develop in this section allows us to talk about *infinite* sequences as purely algebraic objects.

4.1 Ordinary Generating Functions

For a sequence $a_0, a_1, \dots$, denoted $(a_n)_{n \geq 0}$, we define its **ordinary generating function** (OGF) to be

$$A(x) := \sum_{k=0}^{\infty} a_k x^k.$$

In technical terms, this is a power series whose x^i coefficient is a_i . We attach the label “power series” rather than “polynomial” since A need not have a finite degree. Importantly, two OGFs are the same if and only if their corresponding sequences are identical.

Exercise. Let $(a_n)_{n \geq 0}$ and $(b_n)_{n \geq 0}$ have OGFs $A(x)$ and $B(x)$, respectively. Verify the following:

- The OGF for the sequence $(a_n + b_n)_{n \geq 0}$ is $A(x) + B(x)$.
- The OGF $x^k A(x)$ encodes the sequence

$$(\underbrace{0, \dots, 0}_k, a_0, a_1, \dots).$$

- The OGF $A(x)B(x)$ encodes the sequence whose n th term is

$$\sum_{k=0}^n a_k b_{n-k}.$$

- The OGF $xA'(x)$ encodes $(na_n)_{n \geq 0}$.

The above properties allow us to combine sequences in interesting ways. In addition to this, we have the simple geometric series

$$\sum_{k \geq 0} x^k = \frac{1}{1-x},$$

which perhaps the most useful identity in manipulating OGFs. (Question: What is the derivative of this expression, and what is the corresponding power series?)

One of the most useful contexts in which one finds generating functions is the solving of recurrence relations.

Problem (Binet's Formula). Let $f_0 = 0$, $f_1 = 1$, and $f_n = f_{n-1} + f_{n-2}$ for $n \geq 2$ (i.e. f_n is the n th Fibonacci number). Then,

$$f_n = \frac{\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n}{\sqrt{5}}.$$

Proof. Consider the OGF $F(x)$ of (f_n) . Then,

$$\begin{aligned} F(x) &= \sum_{n \geq 0} f_n x^n \\ &= \sum_{n \geq 2} (f_{n-1} + f_{n-2})x^n + f_1 x + f_0 \\ &= x \sum_{n \geq 0} f_n x^n + x^2 \sum_{n \geq 0} f_n x^n + f_1 x + f_0 \\ &= (x + x^2)F(x) + x. \end{aligned}$$

Using partial fraction decomposition,

$$F(x) = \frac{x}{1-x-x^2} = \frac{1}{(1-r_1x)\sqrt{5}} - \frac{1}{(1-r_2x)\sqrt{5}},$$

where $r_1 = \frac{1+\sqrt{5}}{2}$ and $r_2 = \frac{1-\sqrt{5}}{2}$. Using the geometric series formula,

$$F(x) = \sum_{n \geq 0} \frac{(r_1^n - r_2^n)x^n}{\sqrt{5}}$$

which gives the claimed formula for f_n .

4.1.1 Exercises

1. Given any second order linear recurrence $a_n = Pa_{n-1} + Qa_{n-2}$ with $a_0 = c_0$ and $a_1 = c_1$, compute a_k in terms of P , Q , c_0 , c_1 , and k . (Be careful about the case where $x^2 - Px - Q$ has a double root!) Do you see how to extend this to a third order recurrence? An m th order recurrence?

2. Prove that

$$\frac{x^m}{(1-x)^{m+1}} = \binom{k}{m} x^k.$$

3. Recall the binomial theorem from earlier. In the language of generating functions, we can say that $(x+1)^n$ is the OGF for the sequence $a_k = \binom{n}{k}$. However, this can be generalized: in particular, the power series expansion of $(x+1)^r$, for any *real* r , is the OGF for $b_k = \binom{r}{k}$. Find a formula for b_k .

4.2 Exponential Generating Functions

The **exponential generating function** (EGF) of $(a_n)_{n \geq 0}$ is

$$\hat{A}(x) := \sum_{k=0}^{\infty} \frac{a_k}{k!} x^k.$$

Note: The hat above the A is not standard notation, but is used here as a reminder that this is not an OGF.

Much like the crucial identity for manipulating OGFs is

$$\sum_{k \geq 0} x^k = 1 + x + x^2 + x^3 + \cdots = \frac{1}{1-x},$$

the corresponding identity for EGFs is

$$\sum_{k \geq 0} \frac{x^k}{k!} = \frac{1}{0!} + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = e^x.$$

4.2.1 Exercises

1. Show that...

- the EGF for the sequence $0, 1, 2, 3, \dots$ is xe^x .
- the EGF for the sequence $0, 1^2, 2^2, 3^2, \dots$ is $(x+x^2)e^x$.
- the EGF for the sequence $0!, 1!, 2!, 3!, \dots$ is $\frac{1}{1-x}$.
- the EGF for the sequence $\binom{0}{k}, \binom{1}{k}, \binom{2}{k}, \binom{3}{k}, \dots$ is $\frac{x^k}{k!} e^x$.

2. Let $A(x)$ and $B(x)$ be the EGFs of $(a_n)_{n \geq 0}$ and $(b_n)_{n \geq 0}$. Let $A(x)B(x)$ be the generating function for $(c_n)_{n \geq 0}$. Show that

$$c_n = \sum_{k \geq 0} \binom{n}{k} a_k b_{n-k}.$$

Manipulating EGFs is similar to manipulating OGFs, but with a slightly different flavor; there is not much new ground here.

4.3 Digression: What Does x Represent?

Often when working with a function, the point is to somehow map inputs to outputs in a meaningful way. Specifically, both the input and the output have some inherent meaning in the context of the problem.

In this sense, generating functions are not so much functions as they are an organizational tool. In our examples so far, generating functions have been used to sort items by some assigned index. This utility is the reason for the existence of x : it is a dummy variable which, due to its algebraic properties, is perfectly suited for such organizational structure.

Of course, one can derive information by inputting values into a generating function; this is what we did with the sum-of-subsets problem earlier. This is not, however, the goal of a generating function; x still does not have a meaning.

4.4 The Snake Oil Method

The Snake Oil Method is an algebraic trick for evaluating difficult summations. The steps are as follows:

- For some sum $a_n = \sum_k f(k, n)$, consider its OGF

$$A(x) := \sum_{n=0}^{\infty} \left[\sum_k f(k, n) x^n \right].$$

- Swap the order of summation: instead of summing over k , we instead will sum over n , with the hope that this is an easier sum to evaluate. In particular,

$$A(x) = \sum_k \left[\sum_{n \geq 0} f(k, n) x^n \right]$$

- Evaluate $A(x)$, and extract a_n from the coefficients.

The guiding principle behind this trick is that, since addition is commutative, we can swap the order of summation, potentially getting an easier sum.

The power of the Snake Oil Method is best illustrated with an example.

Problem. Evaluate

$$a_n = \sum_{k \geq 0} \binom{k}{n-k}.$$

Solution. Applying Snake Oil, we consider

$$A(x) := \sum_{n \geq 0} a_n = \sum_{n \geq 0} \sum_{k \geq 0} \binom{k}{n-k} x^n.$$

Then, swapping order of summation gives

$$\begin{aligned}
A(x) &= \sum_{k \geq 0} \sum_{n \geq 0} \binom{k}{n-k} x^n \\
&= \sum_{k \geq 0} x^k \sum_{n \geq 0} \binom{k}{n-k} x^{n-k} \\
&= \sum_{k \geq 0} x^k \sum_{r \geq 0} \binom{k}{r} x^r \\
&= \sum_{k \geq 0} x^k (x+1)^k \\
&= \sum_{k \geq 0} (x+x^2)^k \\
&= \frac{1}{1-x-x^2}
\end{aligned}$$

where $r = n-k$. As it turns out, this is (almost) the generating function for the Fibonacci numbers! Thus, the coefficient of x^n in $A(x)$ is $a_n = f_{n+1}$, the $(n+1)$ th Fibonacci number.

Here's another, more convoluted example.

Problem. Prove that

$$\sum_k \binom{m}{k} \binom{n+k}{m} = \sum_k \binom{m}{k} \binom{n}{k} 2^k.$$

Solution. We can use Snake Oil here, too, by showing that both sides have the same OGF. In particular, the generating function for the left side is

$$\begin{aligned}
A(x) &:= \sum_{n \geq 0} \sum_k \binom{m}{k} \binom{n+k}{m} x^n \\
&= \sum_k \sum_{n \geq 0} \binom{m}{k} \binom{n+k}{m} x^n \\
&= \sum_k \binom{m}{k} x^{-k} \sum_{n \geq 0} \binom{n+k}{m} x^{n+k} \\
&= \sum_k \binom{m}{k} x^{-k} \frac{x^m}{(1-x)^{m+1}} \\
&= \frac{x^m}{(1-x)^{m+1}} \cdot \left(1 + \frac{1}{x}\right)^m \\
&= \frac{(1+x)^m}{(1-x)^{m+1}}.
\end{aligned}$$

Doing the same thing on the right hand side gives

$$\begin{aligned}
B(x) &:= \sum_{n \geq 0} \sum_k \binom{m}{k} \binom{n}{k} 2^k x^n \\
&= \sum_k \binom{m}{k} 2^k \sum_{n \geq 0} \binom{n}{k} x^n \\
&= \sum_k \binom{m}{k} \frac{2^k \cdot x^k}{(1-x)^{k+1}} \\
&= \frac{1}{1-x} \sum_k \binom{m}{k} \left(\frac{2x}{1-x} \right)^k \\
&= \frac{1}{1-x} \left(1 + \frac{2x}{1-x} \right)^m \\
&= \frac{(1+x)^m}{(1-x)^{m+1}}.
\end{aligned}$$

Since $A(x) \equiv B(x)$, their corresponding sequences are the same, which is sufficient to prove the problem statement.

4.4.1 Practice Problems

1. For $n \geq 0$, compute

$$\sum_{a+b=n} \binom{2a}{a} \binom{2b}{b}.$$

2. For $m, n \geq 1$, compute

$$\sum_{k \geq 0} \binom{n+k}{m+2k} \binom{2k}{k} \frac{(-1)^k}{k+1}.$$

You may need the identity

$$\sum_{k \geq 0} \frac{1}{k+1} \binom{2k}{k} = \frac{1 - \sqrt{1-4x}}{2x}.$$

5 Counting with Functions, Part 2

5.1 Choosing Without Order

Generating functions are, as mentioned earlier, a way to filter information about a set of objects. The ideas in the previous section further ameliorate our ability to do this, with our newfound ability to talk about infinite sequences.

Operations between ordinary generating functions reflect the structure of choosing things without order. For example:

Problem. In how many ways can we fill a bag with n fruits subject to the following constraints?

- The number of apples must be even.

- The number of bananas must be a multiple of 5.
- There can be at most four oranges.
- There can be at most one pear.

Solution. Again, generating functions filter information. What is the mechanism for filtering here? Well, the obvious sequence we want is T_n , the number of ways to choose n fruits. This motivates filtering by the number of fruits.

Consider the function

$$A(x) := 1 + x^2 + x^4 + \cdots = \frac{1}{1 - x^2}.$$

This encodes the sequence $(1, 0, 1, 0, 1, \dots)$, which is the number of ways to choose n apples. Similarly,

$$B(x) := 1 + x^5 + x^{10} + \cdots = \frac{1}{1 - x^5}$$

encodes the number of ways to choose n bananas.

We also can only choose 0, 1, 2, 3, or 4 oranges, so

$$O(x) := 1 + x + x^2 + x^3 + x^4 = \frac{x^5 - 1}{x - 1},$$

and we can only choose 0 or 1 pears, so

$$P(x) := 1 + x = \frac{x^2 - 1}{x - 1}.$$

The important part is the following theorem:

Theorem. Let $R(x)$ encode r_n , the number of ways to choose n items from $\mathcal{R}$ and $S(x)$ encode s_n , the number of ways to choose n items from $\mathcal{S}$. If $\mathcal{R}$ and $\mathcal{S}$ are disjoint, then $R(x)S(x)$ is the OGF for c_n , the number of ways to choose n items from $\mathcal{R} \sqcup \mathcal{S}$.

Proof. Suppose we take j items from $\mathcal{R}$. We have r_j ways to do this. Then, we must take $n - j$ items from $\mathcal{S}$ to get n total items. This gives a total of $r_j r_{n-j}$ ways to choose n items, j of which are in $\mathcal{R}$. Summing over $j \in \{0, 1, \dots, n\}$ gives

$$c_n = r_0 s_n + r_1 s_{n-1} + \cdots + r_{n-1} s_1 + r_n s_0,$$

which is precisely the x^n coefficient of $R(x)S(x)$.

This theorem can, of course, be extended to more than two sets. In our case, applying it shows us that the OGF for t_n is

$$T(x) = A(x)B(x)O(x)P(x) = \frac{1}{1 - x^2} \cdot \frac{1}{1 - x^5} \cdot \frac{x^5 - 1}{x - 1} \cdot \frac{x^2 - 1}{x - 1} = \frac{1}{(1 - x)^2},$$

which we find, upon differentiating $\frac{1}{1-x}$, turns out to encode the sequence $t_n = n + 1$.

Exercise. Make your own problem! Try imposing arbitrary restrictions on the fruits and finding the corresponding generating function for t_n .

Here's another nice example!

Problem. To *triangulate* a k -sided polygon, we draw $k - 2$ non-intersecting diagonals, forming $k - 2$ triangles. How many ways are there to triangulate a k -sided polygon?

Solution. The key insight is that we can generate a recursion due to the inherent self-similarity of the setup. Let C_{k-2} be the number of triangulations of a k -gon.

Label the vertices $1, 2, \dots, k$ in order, and let (a, b) denote the segment connecting a and b . Then, $(1, k)$ must be a part of some triangle in this triangulation, whose third vertex is at i . Of course, $2 \leq i \leq k - 1$. Segments $(1, i)$ and (k, i) create two smaller polygons: one with i sides and one with $k + 1 - i$ sides. Hence, the number of ways to triangulate the new polygons is $C_{i-2}C_{k-1-i}$.

Summing over all i , we get that

$$C_{k-2} = \sum_{i=2}^{k-1} C_{i-2}C_{k-1-i},$$

or equivalently,

$$C_k = \sum_{i=0}^{k-1} C_i C_{k-i-1}$$

after re-indexing.

Before continuing, it's worth looking at the structure of this recurrence: importantly, the expression for C_k looks a lot like the result of squaring an OGF, since it is just

$$\sum_{i=0}^{k-1} a_i b_{k-1-i}$$

where we take both (a_k) and (b_k) to be the sequence (C_k) .

This is the final push that we need, since, taking $C(x)$ to be the OGF of (C_k) gives us the identity

$$G(x)^2 = 1 + xG(x)^2.$$

Funnily enough, this is a quadratic in $G(x)$, whose solutions are

$$G(x) = \frac{1 \pm \sqrt{1 - 4x}}{2x}.$$

We now only need to decide which of the solutions is correct. We get this by taking $x = 0$, since $G(0) = C_0 = 1$. Analyzing both solutions shows us that, in fact,

$$G(x) = \frac{1 - \sqrt{1 - 4x}}{2x}.$$

Exercise. The generalized binomial theorem says that

$$(1+x)^r = \sum_{n \geq 0} \binom{r}{n} x^n$$

for any real r . Use this to find a closed form for C_k . You will need the fact that

$$\binom{r}{n} = \frac{r(r-1) \cdots (r-n+1)}{n!}.$$

Fun fact: The numbers of the sequence (C_k) are called the **Catalan numbers**.

5.2 Choosing With Order

Just as OGFs are good for unordered choosing, EGFs work well for ordered choosing. This is due to the following theorem:

Theorem. Consider n objects in a line. Let a_k be the number of ways to give property α to a sequence of k objects, and b_k be the number of ways to give property β to a sequence of k objects. Let c_n be the number of ways to assign exactly one of properties α and β to the n objects. Then, if $A(x)$ and $B(x)$ are the EGFs of a_k and b_k , the function $A(x)B(x)$ is the EGF of (c_n) .

Solution. We know that the x^n coefficient of $A(x)B(x)$ is

$$\sum_{k \geq 0} \binom{n}{k} a_k b_{n-k}.$$

The goal is to show that this counts what we want. Fix k objects to give property α to; there are $\binom{n}{k}$ ways to choose these k . Then, there are $a_k b_{n-k}$ ways to assign the properties themselves. Thus, there are

$$\binom{n}{k} a_k b_{n-k}$$

ways to assign property α to any subset of k objects. Summing over all k gives the desired result.

Here's a pretty straightforward application of the theorem.

Problem. How many words are there in the alphabet $\{a, b, c, d\}$ such that there are an even number of as and bs (total) and at most 8 cs ?

Solution. Let t_n be the number of such words of length n . Our strategy is to construct an EGF for the number of as and bs together, and then multiply it by the EGFs for the number of cs and ds .

For the as and bs , take an even number $2m$. There are 2^{2m} ways to write a string of $2m$ as or bs , so the EGF for this would then be

$$F(x) := \sum_{k \geq 0} \frac{(2x)^{2k}}{(2k)!} = \frac{1}{2}(e^{2x} + e^{-2x}).$$

Since we have at most 8 cs , the EGF for cs is

$$C(x) := \sum_{k=0}^8 \frac{x^k}{k!}.$$

Finally, since ds are not restricted, the EGF for the number of ds is $D(x) := e^x$. The product is thus

$$\frac{1}{2}(e^{2x} + e^{-2x}) \cdot \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \cdots + \frac{x^8}{8!}\right) \cdot e^x.$$

To find a closed form, we rewrite the e^x terms to get

$$C(x) \sum_{k \geq 0} \frac{3^k + (-1)^k}{2} \cdot \frac{x^k}{k!},$$

so that the x^n coefficient is

$$\sum_{k=0}^8 \binom{n}{k} \left(\frac{3^k + (-1)^k}{2} \right),$$

where the summation only goes up to 8 because $C(x)$ only has nonzero coefficients up to x^8 , and this is the desired answer, however messy it may seem.

Exercise. Make your own problem, part 2! Perturb the setup, and find the corresponding generating function.

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Elements of Mathematical Linguistics

Shaon Anwar



1 Introduction

It is no coincidence that many linguistic researchers have a background in mathematics. As the scientific study of language, linguistics requires well-defined mathematical models, and the mathematical precision of language is indispensable in testing and implementing theories of language. Without the helping hand of mathematics, the crucial questions of the extensive grammars in computational applications of language processing cannot be addressed. Since language is structured at several levels (phonology, morphology, syntax, and semantics), mathematical concepts and techniques have been developed and applied to each area, thus forming subdivisions in mathematical linguistics.

2 Probability Meets Poetry

In January 1913, Andrei Markov perused the first 20,000 words of Pushkin's Eugene Onegin, painstakingly counting vowels (8,638) and consonants (11,362) and noting the order of their appearance as vowel-vowel pairs (1,104), consonant-consonant pairs (3,827) and combinations of the two speech sounds (15,069). Interestingly enough, his analysis started with the goal of rebuffing another mathematician, Pavel Nekrasov, with whom he had disputed for years. Specifically, Nekrasov's claim (rooted in his theological belief of free will) that the Weak Law of Large Numbers (the idea that as a sample size becomes larger, the sample mean gets closer to the expected value) applied only to independent variables, set Markov on the inquiry to prove Nekrasov's assertion incorrect. At the end, Markov would show that the Law of Large Numbers applies to systems of dependent variables.

Markov divided the whole 20,000 letter sequence into 200 separate sequences of 100 letters, which were arranged into 10 by 10 tables. He then counted the vowels in each column of the table, and added the numbers of vowels from the first and sixth columns, second and seventh columns, third and eighth columns, etc. (Those sums are written vertically as seen below.) Finally, he combined each vertically written number. (We see the final sums as written in bold.) The most frequent value of vowels among the listed sums was 43 (as seen below). Markov calculated the mean number of vowels per block of 100 letters to be approximately 43, and the variance to be 2.26. Independence of chunks of letters (one 100-letter sequence had no effect on the next 100-letter sequence) and randomness of the sample followed a normal distribution.

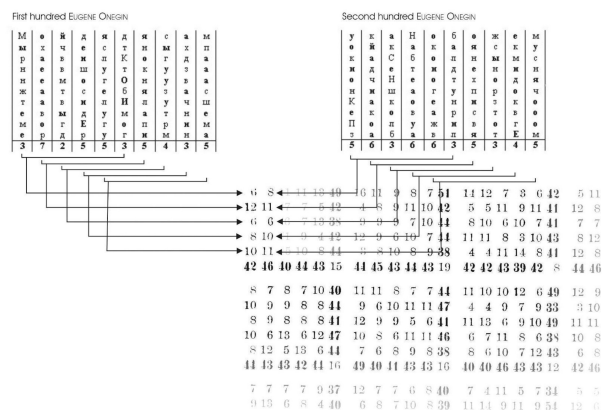


Figure 1: Markov's counting method

Next, he turned to calculate the probability of two vowels in succession. Assuming two adjacent letters were independent, the probability would be equal to $0.43 * 0.43$, or about 0.19. In such a case, our sample of letters would be expected to have about 3,800 vowel pairs out of the 20,000 letters. This count, however, exceeded the actual number of the vowel-vowel pairings appearing in the text (1,104) by threefold. In addition, given phonetic features of language, there exists a strong tendency for alternating vowels and consonants. Hence, there was an evident dependency among letters; in a sequence or “chain” of letters, each letter was dependent on the preceding one. For example, a certain batch of consonants forced the next letter to be a vowel. Such events, linking one to the next, formed what came to be known as a Markov chain, in which the future state of the system depended only on the current state. Markov chains can be represented as probabilistic state diagrams where the transitions between events or states are described with the probabilities of their occurrences.

3 Shannon and Language Modeling

Markov's investigation into probabilistic letter sequences found its way into Claude Shannon's 1948 paper, “A Mathematical Theory of Communication,” and was used as a mathematical model to predict the sequences of words in a piece of English text. First, Shannon used a Markov model of order 0 to make a prediction of how likely a given letter occurs in the texts; in other words, based on its frequency, he would predict a fixed probability of the letter. For example, in the text “abcbbaacc,” the predictability of ‘a’ occurring is $2/9$, ‘b’ $3/9$ and ‘c’ $4/9$. According to this model, based on independence assumption and given probability, any stream of letters could be effectively generated. However, as groups of letters in the English language occur together with certain degrees

of correlation ('t' is more often followed by 'h' and not 'x'), Shannon expanded this Markov model by relating the probability of a chosen letter to its successive or preceding letters. Then, Shannon explored the idea of higher-order Markov models: bigrams (pairs of words), trigrams (three successive words), and n -grams (a sequence of n words) with each state dependent on 2, 3 or n prior steps.

He thought that higher order Markov models would get us closer to understanding the fundamental concepts of language. However, with the rise of the Chomskyan approach to human grammatical knowledge and the argument that Markov chains did not capture relationships between words, research into application of probability theory to linguistics stagnated for decades.

Before discussing the grammatical approach to linguistics, let's turn to the notion of sets in the domain of language.

4 Notation

A mathematical concept such as set theory is important in language study because language can be viewed, on a basic level, as a collection of elements such as strings or words which, when stitched together, form sentences. Hence, just as sets, sentence structures are abstract constructs containing elements that follow certain descriptions. Let's start with strings, which are fundamental structures of language. (We cannot read this article without understanding their workings.) An alphabet, denoted as " Σ ," can be defined as a finite set of symbols. Any string created from an alphabet is a finite set of symbols of Σ . Any set of such strings is referred to as a formal language over an alphabet. For example, in an alphabet Σ with symbols a, b, c, a certain language must have strings that start with symbol b and contain two characters. This means that the possible strings for this language include "ba," "bb," and "bc."

A language can be further characterized as " Σ^n " (the set of strings with a length of n), " Σ^+ " (the set of strings with a length of 1 or greater, also known as the positive closure of Σ), and " Σ^* " (the set of all strings, also known as the Kleene closure or Σ).

5 Chomsky and "colorless green ideas sleep furiously"

In the 1950s, in an attempt to understand the principles that govern human speech, linguist Noam Chomsky defined a set of rules called grammars, which specified the structural description of each sentence. The sentence "colorless green ideas sleep furiously" follows the rules of syntax, yet it is meaningless; grouping words in a linear order does not sufficiently capture the unit's meaning. It illustrates Chomsky's point that the probabilistic approach to syntax takes us only halfway to the goal of generating language to convey meaning.

Using grammar structures such as

$$\begin{aligned} \text{DNP VP} &\rightarrow \text{S} \\ \text{V DNP} &\rightarrow \text{VP} \\ \text{P DNP} &\rightarrow \text{PP}, \end{aligned}$$

where

- DNP = definite noun phrase

- VP = verb phrase
- S = sentence
- V = verb
- P = preposition
- PP = prepositional phrase

and a parse tree, Chomsky broke down human speech patterns as each word corresponded to a given type.

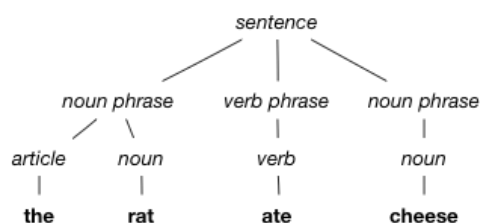


Figure 2: Parse tree of a simple sentence

This hierarchical structure of language is used extensively in natural language processing (NLP), which is the concept of artificial comprehension of the human language. The process of breaking down sentences into their components, known as tokenization, allows for machines to understand (or not understand, in the case of a nonsensical phrase) written language.

Another such implementation of language hierarchy is in the finite-state automata, which is a machine that monitors and changes state based on inputs. Upon processing each input, the machine adjusts for the most likely successive input. For example, a finite-state machine can process alphabetical inputs and predict phrases by utilizing logical grammar structures based on parse trees.

6 Conclusion

Overall, the goal of mathematical linguistics is to gain a better understanding of the nature of human language and how it is processed by the brain. Over the decades, mathematical methods to analyze multiple structural levels of language have been developed, changed and adopted. Markov's probabilistic sequential text analysis proved to be a momentous, early example of the application of mathematical techniques to the study of language, and it paved the way for further research in the field. Incorporating Markov's models, Shannon's work gave a greater insight into the relationship between successive vowels and consonants in forming words and sentences. (A common application of Markov chains is text prediction; when users write emails, Google predicts words to autocomplete it or suggests replies to received messages.) In the 1950s, there was a gradual shift from statistical modeling to theoretical foundations where attention was paid to formal logical approaches to language. However, in the 1970s, enabled by increasing availability of machine-readable text data, the probability theory in linguistics reemerged, and since then has produced more sophisticated probabilistic models. Currently, the corpus-based statistical training and parsing techniques are used

in conjunction with modern language modeling done with neural network language models. This development in finding and using mathematical models to solve linguistic problems will continue, thanks to larger and more accessible linguistic data, increased research in AI, and the processing power of modern machines.

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Shapiro's Inequality: A Remarkable Use of Convex Functions

Daniel Potievsky



1 Introduction

This article discusses an interesting inequality conjecture that was originally published by the American mathematician Harold Shapiro in *The American Mathematical Monthly* in 1954. It attracted the attention of many renowned mathematicians around the world and was eventually resolved by Vladimir Drinfeld, the 1990 Fields Medal winner, at 17 years old. The statement of the inequality is as follows:

$$\frac{x_1}{x_2 + x_3} + \frac{x_2}{x_3 + x_4} + \dots + \frac{x_{n-1}}{x_n + x_1} + \frac{x_n}{x_1 + x_2} \geq \frac{n}{2}, \quad x_i > 0.$$

2 The Early Cases

The story of this inequality begins much earlier, with the publication of Nesbitt's Inequality in 1902. Nesbitt's Inequality states that for positive numbers a , b , and c ,

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

This is simply the case of $n = 3$ for the Shapiro Inequality. Over the years, many proofs have been developed for this inequality, and perhaps one of the most insightful is through Jensen's Inequality for convex functions. This inequality states that if a function Φ is convex in some neighborhood I ,

then

$$\lambda_1\Phi(x_1) + \lambda_2\Phi(x_2) + \dots + \lambda_n\Phi(x_n) \geq \Phi(\lambda_1x_1 + \lambda_2x_2 + \dots + \lambda_nx_n)$$

when $\lambda_1 + \lambda_2 + \dots + \lambda_n = 1$ and $x_1, x_2, \dots, x_n \in I$.

2.1 Nesbitt's Inequality

First, consider the proof of Nesbitt's Inequality:

Let $a + b + c = 1$. Fixing the sum in this manner is permissible, because this inequality is homogeneous, meaning that if we multiply each of a , b , and c by some constant ζ , then

$$\frac{\zeta a}{\zeta b + \zeta c} + \frac{\zeta b}{\zeta c + \zeta a} + \frac{\zeta c}{\zeta a + \zeta b} = \frac{a}{b + c} + \frac{b}{c + a} + \frac{c}{a + b},$$

and the inequality does not change, since ζ can be divided out. Therefore, if we let the sum $a + b + c = S$ for some arbitrary S , then we can always let $\zeta = S$, which does not change the inequality but permits $Sa + Sb + Sc = S$ or $a + b + c = 1$.

Therefore, we obtain that $\frac{a}{b + c} + \frac{b}{c + a} + \frac{c}{a + b} = \frac{a}{1 - a} + \frac{b}{1 - b} + \frac{c}{1 - c}$.

Let $f(x) = \frac{1}{1-x}$. Then, $f''(x) = \frac{2}{(1-x)^3}$. By examining where $f''(x) \geq 0$, we can conclude that $f(x)$ is convex on the interval $(0, 1)$. But from the condition of $a + b + c = 1$ and the condition that $a, b, c > 0$, we can assert that $a, b, c \in (0, 1)$. So, we can use Jensen's Inequality.

Note that

$$\frac{\frac{a}{1-a} + \frac{b}{1-b} + \frac{c}{1-c}}{3} = \frac{1}{3} \cdot \frac{a}{1-a} + \frac{1}{3} \cdot \frac{b}{1-b} + \frac{1}{3} \cdot \frac{c}{1-c} \geq \frac{\frac{a}{3} + \frac{b}{3} + \frac{c}{3}}{1 - (\frac{a}{3} + \frac{b}{3} + \frac{c}{3})} = \frac{1}{2}$$

by Jensen's Inequality. Therefore,

$$\frac{\frac{a}{1-a} + \frac{b}{1-b} + \frac{c}{1-c}}{3} \geq \frac{1}{2},$$

which means

$$\frac{a}{1-a} + \frac{b}{1-b} + \frac{c}{1-c} = \frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2},$$

completing the proof.

2.2 Mordell's Inequality

But the fight for a generalization of this inequality continued. In 1958, the Cambridge mathematician Louis Mordell managed to prove the corresponding inequalities for 4, 5, and 6 variables using a more refined technique with Jensen's Inequality. This technique can be illustrated with the case of 4 variables:

2.2.1 Claim

$$\frac{a}{b+c} + \frac{b}{c+d} + \frac{c}{d+a} + \frac{d}{a+b} \geq 2 \text{ for positive reals } a, b, c, \text{ and } d.$$

2.2.2 Proof

First, let us write out Jensen's Inequality for $f(x) = \frac{1}{x}$ (since $f''(x) = \frac{2}{x^3}$, this function is convex over the set of all positive reals). We obtain:

$$\lambda_1 x_1^{-1} + \lambda_2 x_2^{-1} + \dots + \lambda_n x_n^{-1} \geq \frac{1}{\lambda_1 x_1 + \lambda_2 x_2 + \dots + \lambda_n x_n}.$$

But since $\lambda_1 + \lambda_2 + \dots + \lambda_n = 1$ by the condition for Jensen's Inequality, we can rewrite this as

$$\frac{\lambda_1 x_1^{-1} + \lambda_2 x_2^{-1} + \dots + \lambda_n x_n^{-1}}{\lambda_1 + \lambda_2 + \dots + \lambda_n} \geq \frac{\lambda_1 + \lambda_2 + \dots + \lambda_n}{\lambda_1 x_1 + \lambda_2 x_2 + \dots + \lambda_n x_n}$$

or

$$\frac{\sum \lambda_i x_i^{-1}}{\sum \lambda_i} \geq \frac{\sum \lambda_i}{\sum \lambda_i x_i}.$$

From here, we obtain that

$$\sum \lambda_i x_i^{-1} \geq \frac{(\sum \lambda_i)^2}{\sum \lambda_i x_i}.$$

Returning to the problem with four variables, make the following substitutions: $\lambda_1 = a$, $\lambda_2 = b$, $\lambda_3 = c$, $\lambda_4 = d$. Once again, this is possible because the inequality is homogeneous and we can always say that $a + b + c + d = 1$. We can also let $x_1 = b + c$, $x_2 = c + d$, $x_3 = d + a$, $x_4 = a + b$. Using the inequality that was derived from Jensen's Inequality for four variables, we get

$$\frac{a}{b+c} + \frac{b}{c+d} + \frac{c}{d+a} + \frac{d}{a+b} \geq \frac{(a+b+c+d)^2}{a(b+c) + b(c+d) + c(d+a) + d(a+b)}.$$

Therefore, it is sufficient to prove that

$$\frac{(a+b+c+d)^2}{a(b+c) + b(c+d) + c(d+a) + d(a+b)} \geq 2.$$

Opening parentheses and clearing denominators, we obtain:

$$\begin{aligned} (a+b+c+d)^2 &\geq 2(ab+2ac+ad+bc+2bd+cd) \\ (a^2+b^2+c^2+d^2) + 2(ab+ac+ad+bc+bd+cd) &\geq 2(ab+2ac+ad+bc+2bd+cd) \\ (a^2+b^2+c^2+d^2) &\geq 2ac+2bd \\ (a-c)^2 + (b-d)^2 &\geq 0, \end{aligned}$$

which is clearly true.

2.3 Exercise

Prove the analogous statements for five and six variables.

Hint: The final step will require the Cauchy-Schwarz Inequality.

3 Theorems

Unfortunately, the problem began for $n = 7$, which did not yield to Mordell's technique. Over the years, many attempts were made to resolve the Shapiro Inequality, and there was a certain degree of success achieved with individual cases. In 1961, the case $n = 8$ was resolved; in 1967, the case $n = 10$ was resolved; and in 1974, the case $n = 12$ was resolved. But the questions of whether or not the inequality was true for all n and whether the conjecture was right at all remained.

It turned out that the cyclic nature of this inequality was a great help for conducting further proofs. Denote for simplicity

$$f_n(x_1, x_2, \dots, x_n) = \frac{x_1}{x_2 + x_3} + \frac{x_2}{x_3 + x_4} + \dots + \frac{x_{n-1}}{x_n + x_1} + \frac{x_n}{x_1 + x_2}.$$

We can see by inspecting that this expression is cyclic, because

$$f_n(x_1, x_2, \dots, x_n) = f_n(x_k, x_{k+1}, \dots, x_n, x_1, x_2, \dots, x_{k-1}).$$

3.1 Theorem 1

3.1.1 Claim

If Shapiro's Inequality is untrue for m variables, then it is untrue for $m + 2$ variables.

3.1.2 Proof

Suppose for some m , $f_m(x_1, x_2, \dots, x_m) < \frac{m}{2}$. One can then construct a set of $m + 2$ variables such that $f_m(x_1, x_2, \dots, x_{m+2}) < \frac{m+2}{2}$ as well, which completes the proof. For the proof, use

$$\begin{aligned} f_{m+2}(x_1, x_2, x_3, \dots, x_m, x_1, x_2) &= \frac{x_1}{x_2 + x_3} + \frac{x_2}{x_3 + x_4} + \dots + \frac{x_m}{x_1 + x_2} + \frac{x_1}{x_2 + x_1} + \frac{x_2}{x_1 + x_2} \\ &= f_m(x_1, x_2, \dots, x_m) + \frac{x_1}{x_1 + x_2} + \frac{x_2}{x_1 + x_2} \\ &= f_m(x_1, x_2, \dots, x_m) + 1. \end{aligned}$$

Therefore,

$$f_{m+2}(x_1, x_2, x_3, \dots, x_m, x_1, x_2) = f_m(x_1, x_2, \dots, x_m) + 1 < \frac{m}{2} + 1 = \frac{m+2}{2},$$

or

$$f_{m+2}(x_1, x_2, x_3, \dots, x_m, x_1, x_2) < \frac{m+2}{2},$$

which concludes the proof.

3.2 Theorem 2

3.2.1 Claim

If Shapiro's Inequality is untrue for m variables, where m is an odd integer, then it is untrue for all integers greater than m as well.

3.2.2 Proof

First note the following identity:

$$f_{m+1}(x_1, x_2, \dots, x_i, x_i, \dots, x_m, x_{m+1}) - f_m(x_1, x_2, \dots, x_m) - \frac{1}{2} = \frac{(x_i - x_{i-1})(x_i - x_{i+1})}{2x_i(x_i + x_{i+1})}.$$

This is true because the left side can be rewritten as

$$\begin{aligned} & \left(\frac{x_1}{x_2 + x_3} + \dots + \frac{x_{i-1}}{x_i + x_i} + \frac{x_i}{x_i + x_{i+1}} + \frac{x_i}{x_{i+1} + x_{i+2}} + \dots + \frac{x_m}{x_1 + x_2} \right) - \left(\frac{x_1}{x_2 + x_3} + \dots + \frac{x_m}{x_1 + x_2} \right) - \frac{1}{2} \\ &= \frac{x_{i-1}}{x_i + x_i} + \frac{x_i}{x_i + x_{i+1}} - \frac{x_{i-1}}{x_i + x_{i+1}} - \frac{1}{2}, \end{aligned}$$

with the rest of the terms cancelling. Bringing to a common denominator, this expression equals

$$\begin{aligned} \frac{x_{i-1}(x_i + x_{i+1}) + 2x_i^2 - 2x_{i-1}x_i - x_i(x_i + x_{i+1})}{2x_i(x_i + x_{i+1})} &= \frac{x_{i-1}x_{i+1} + x_i^2 - x_{i-1}x_i - x_ix_{i+1}}{2x_i(x_i + x_{i+1})} \\ &= \frac{(x_i - x_{i-1})(x_i - x_{i+1})}{2x_i(x_i + x_{i+1})}, \end{aligned}$$

as needed.

But in the case that m is odd, note that it is impossible for $(x_i - x_{i-1})(x_i - x_{i+1}) > 0$ for all i . This can be shown by contradiction. If the previous claim were true, then

$$\begin{cases} (x_2 - x_1)(x_2 - x_3) > 0, \\ (x_3 - x_2)(x_3 - x_4) > 0, \\ \vdots \\ (x_m - x_{m-1})(x_m - x_1) > 0, \\ (x_1 - x_m)(x_1 - x_2) > 0. \end{cases}$$

By pairing up the terms that are negatives of each other, multiplying these inequalities out yields that $-(x_1 + x_2)^2(-(x_2 + x_3)^2)\dots(-(x_m + x_1)^2) > 0$. But since there are m parentheses and m is odd, this yields that a negative number is greater than 0, which is a contradiction. Therefore, if m is odd, then there must exist such an i that $(x_i - x_{i-1})(x_i - x_{i+1}) \leq 0$. But using the identity derived earlier, that would mean that

$$f_{m+1}(x_1, x_2, \dots, x_i, x_i, \dots, x_m, x_{m+1}) - f_m(x_1, x_2, \dots, x_m) - \frac{1}{2} \leq 0,$$

or

$$f_{m+1}(x_1, x_2, \dots, x_i, x_i, \dots, x_m, x_{m+1}) \leq f_m(x_1, x_2, \dots, x_m) + \frac{1}{2} < \frac{m+1}{2}$$

by the problem statement. But this actually concludes the proof. If the inequality is untrue for some odd m , then it is also untrue for the even number $m+1$. Using Theorem 1, we can conclude, with an induction over the evens and odds, that the inequality is untrue for all integers greater than m .

4 Resolution

The two previous theorems, though simple, were pivotal to the resolution of the problem. In 1971, for instance, a counterexample was found for the case of $n = 25$. By Theorems 1 and 2, this implies that the inequality is wrong for all $n \geq 25$. In 1985, the mathematician Troesch found a counterexample for $n = 14$, and in 1989, he published a paper that showed that the inequality was true for all odd values up to 23. Thus, in some sense, with much manual work, Shapiro's Inequality was resolved to work only for even values less than or equal to 12 and odd values less than or equal to 23. But the difficulty that mathematicians had with finding counterexamples was very curious, and Shapiro's conjecture of $\frac{n}{2}$ clearly worked for many of the smaller cases. Perhaps the problem needed to be modified.

5 The Question of a Coefficient

In particular, the question arose as to whether the Shapiro Inequality was always true for some coefficient γ , namely:

$$\frac{x_1}{x_2 + x_3} + \frac{x_2}{x_3 + x_4} + \dots + \frac{x_{n-1}}{x_n + x_1} + \frac{x_n}{x_1 + x_2} \geq \gamma \cdot \frac{n}{2}$$

The process of experimenting and seeking out the value of γ lasted many years.

In fact, the following problem appeared in the Third Soviet Mathematics Olympiad in 1969:

5.0.1 Claim

Prove that if all $x_i > 0$, then $\frac{x_1}{x_2 + x_3} + \frac{x_2}{x_3 + x_4} + \dots + \frac{x_{n-1}}{x_n + x_1} + \frac{x_n}{x_1 + x_2} \geq \frac{n}{4}$.

5.0.2 Proof

This inequality is cyclic, so we can assume without loss of generality that x_1 is the maximum of all the x_i . Denote that maximum to be a_1 . Then, implement the following algorithm: look in the denominator of that fraction and find the larger of the two numbers there (x_2 and x_3). Select that maximum number, and denote it a_2 . Then, find the fraction that has this maximum in its numerator. (This will be $\frac{x_2}{x_3 + x_4}$ or $\frac{x_3}{x_4 + x_5}$, depending on whether x_2 or x_3 is larger.) Look in the denominator of that fraction, and denote the maximum of the two numbers there as a_3 , and so on.

Continue this process until step k : reaching an a_k such that the maximum of the next two numbers is $x_1 = a_1$. How is it known that the algorithm will reach x_1 eventually? Looking at the algorithm in action will show that it moves either by 1 or 2 fractions at a time, since after selecting a fraction with x_i in the numerator, the next fraction that will be selected will either have x_{i+1} or x_{i+2} in the numerator. Since x_1 is the maximum of all the x_i and it is contained in the denominator of two consecutive fractions, $\frac{x_{n-1}}{x_n + x_1}$ and $\frac{x_n}{x_1 + x_2}$, the algorithm will not miss it, since it moves by at most 2 fractions. And because the algorithm moves by at most 2 fractions, it can be asserted that $k \geq \frac{n}{2}$, since the minimal number of steps that will complete this algorithm occurs when the step of the algorithm jumps by 2 fractions, not 1, every time.

As such, it can be stated that

$$\frac{x_1}{x_2 + x_3} + \frac{x_2}{x_3 + x_4} + \dots + \frac{x_{n-1}}{x_n + x_1} + \frac{x_n}{x_1 + x_2} \geq \frac{a_1}{2a_2} + \frac{a_2}{2a_3} + \dots + \frac{a_k}{2a_1}.$$

From here, we use the Inequality of Arithmetic and Geometric Means (AM-GM) to get

$$\frac{a_1}{2a_2} + \frac{a_2}{2a_3} + \dots + \frac{a_k}{2a_1} \geq k \cdot \sqrt[k]{\frac{a_1}{2a_2} \cdot \frac{a_2}{2a_3} \cdot \dots \cdot \frac{a_k}{2a_1}} = \frac{k}{2} \geq \frac{n}{4}.$$

The proof is complete. (This is essentially a proof for $\gamma = \frac{1}{2}$.)

5.1 Exercises

1. Prove that $f_n(x_1, x_2, \dots, x_n) \geq (\sqrt{2} - 1)n \approx 0.41n$.

2. Prove that $f_n(x_1, x_2, \dots, x_n) \geq \frac{5}{12}n$.

Hint: Use the identity $\frac{x_i}{x_{i+1}+x_{i+2}} + \alpha = \frac{x_i+\beta x_{i+1}}{x_{i+1}+x_{i+2}} + \frac{\alpha(\beta x_{i+1}+x_{i+2})}{x_{i+1}+x_{i+2}}$, where $\beta + \alpha\beta = \alpha$, or $\beta = \frac{\alpha}{\alpha+1}$. Proceed with a manipulation using AM-GM. The exercises require 2 different values of α .

6 Drinfeld's Final Resolution to the Shapiro Inequality

Suppose $\frac{x_1}{x_2} = k_1, \frac{x_2}{x_3} = k_2, \dots, \frac{x_n}{x_1} = k_n$. Note that $k_1 k_2 \dots k_n = 1$. Take all the values k_i and put them into increasing order. Denote the largest as m_1 , the second as m_2 , $\dots$, the smallest as m_n (note: $m_1 m_2 \dots m_n = 1$). Therefore, we can see that the sequences $\{\frac{1}{m_1}, \frac{1}{m_2}, \dots, \frac{1}{m_n}\}$ and $\{\frac{1}{m_1+1}, \frac{1}{m_2+1}, \dots, \frac{1}{m_n+1}\}$ are non-decreasing; that is, $\frac{1}{m_1} \leq \frac{1}{m_2} \leq \dots \leq \frac{1}{m_n}$ and $\frac{1}{m_1+1} \leq \frac{1}{m_2+1} \leq \dots \leq \frac{1}{m_n+1}$. By the rearrangement inequality, it is known that a sum of pairwise products of two ordered sets $a_1 \geq a_2 \geq \dots \geq a_n$ and $b_1 \geq b_2 \geq \dots \geq b_n$ reaches its minimum when it is arranged as $a_1 b_n + a_2 b_{n-1} + \dots + a_{n-1} b_2 + a_n b_1$.

Therefore,

$$\begin{aligned} \frac{x_1}{x_2 + x_3} + \frac{x_2}{x_3 + x_4} + \dots + \frac{x_{n-1}}{x_1 + x_2} + \frac{x_n}{x_1 + x_2} &= \frac{1}{k_1(k_2 + 1)} + \frac{1}{k_2(k_3 + 1)} + \dots + \frac{1}{k_n(k_1 + 1)} \\ &\geq \frac{1}{m_1(m_n + 1)} + \frac{1}{m_2(m_{n-1} + 1)} + \dots + \frac{1}{m_n(m_1 + 1)}, \end{aligned}$$

or

$$f_n \geq \frac{1}{m_1(m_n + 1)} + \frac{1}{m_2(m_{n-1} + 1)} + \dots + \frac{1}{m_n(m_1 + 1)}$$

where the inequality is obtained from the minimality of the rearrangement inequality for the two ordered sets given. Denote then $r_i = \frac{1}{m_i(m_{n-i+1}+1)} + \frac{1}{m_{n-i+1}(m_i+1)}$. With this definition, we obtain an equivalent inequality:

$$f_n \geq \frac{1}{2} \sum_{i=1}^n r_i$$

Let $c_i = m_i \cdot m_{n-i+1}$. Note that $c_1 c_2 \dots c_n = 1$.

Rewriting r_i , we get

$$\begin{aligned}
r_i &= \frac{m_{n-i+1}(m_i + 1) + m_i(m_{n-i+1} + 1)}{m_i m_{n-i+1}(m_i + 1)(m_{n-i+1} + 1)} \\
&= \frac{2c_i + m_i + m_{n-i+1}}{c_i(m_i + 1)(m_{n-i+1} + 1)} \\
&= \frac{1}{c_i} \left(\frac{2c_i + m_i + m_{n-i+1}}{(m_i + 1)(m_{n-i+1} + 1)} \right) \\
&= \frac{1}{c_i} \left(\frac{c_i + m_i \cdot m_{n-i+1} + m_i + m_{n-i+1}}{(m_i + 1)(m_{n-i+1} + 1)} \right) \\
&= \frac{1}{c_i} \left(\frac{c_i - 1 + m_i \cdot m_{n-i+1} + m_i + m_{n-i+1} + 1}{(m_i + 1)(m_{n-i+1} + 1)} \right) \\
&= \frac{1}{c_i} \left(\frac{c_i - 1}{(m_i + 1)(m_{n-i+1} + 1)} + 1 \right).
\end{aligned}$$

By AM-GM,

$$\begin{aligned}
\frac{m_i + m_{n-i+1}}{2} &\geq \sqrt{m_i \cdot m_{n-i+1}} = \sqrt{c_i} \\
m_i + m_{n-i+1} &\geq 2\sqrt{c_i}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
m_i \cdot m_{n-i+1} + m_i + m_{n-i+1} + 1 &\geq c_i + 2\sqrt{c_i} + 1 \\
(m_i + 1)(m_{n-i+1} + 1) &\geq (1 + \sqrt{c_i})^2.
\end{aligned}$$

If $c_i < 1$, then $\frac{c_i - 1}{(m_i + 1)(m_{n-i+1} + 1)} < 0$, and this value will attain a maximum when the denominator is maximized. Therefore,

$$r_i = \frac{1}{c_i} \left(\frac{c_i - 1}{(m_i + 1)(m_{n-i+1} + 1)} + 1 \right) \geq \frac{1}{c_i} \left(\frac{c_i - 1}{(1 + \sqrt{c_i})^2} + 1 \right) = \frac{2}{c_i + \sqrt{c_i}}$$

when $c_i < 1$. If $c_i \geq 1$, then

$$r_i = \frac{1}{c_i} \left(\frac{c_i - 1}{(m_i + 1)(m_{n-i+1} + 1)} + 1 \right) \geq \frac{1}{c_i}.$$

From the definition of c_i , it follows that $c_i > 0$, because $m_i, m_{n-i+1} > 0$. Then, the following substitution can be made: $c_i = e^{\eta_i}$. Note that $\eta_1 + \eta_2 + \dots + \eta_n = \ln(c_1 c_2 \dots c_n) = 0$. Therefore, from the previous inequalities, we get that

$$f_n \geq \sum_{c_i \geq 1} e^{-\eta_i} + \sum_{c_i < 1} 2(e^{\frac{\eta_i}{2}} + e^{\eta_i})^{-1}.$$

Let $\psi(x)$ be the largest convex function that does not exceed the functions $\mu(x) = e^{-x}$ and $\theta(x) = 2(e^{\frac{x}{2}} + e^x)^{-1}$.

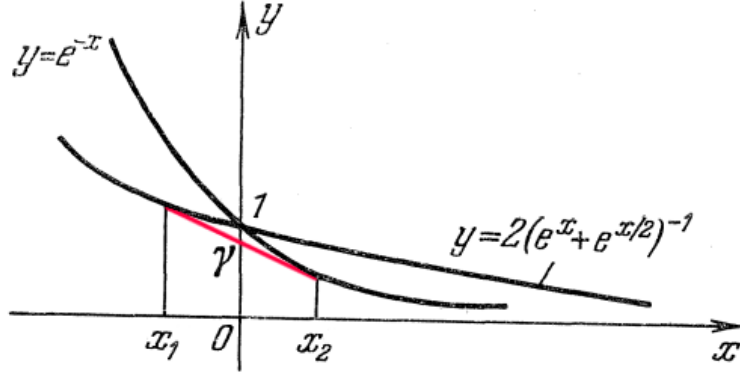


Figure 1: Depiction of $\psi(x)$.

Therefore, assuming that $f_n \geq \gamma \cdot \frac{n}{2}$, we can search for γ by examining $\frac{2}{n}f_n$. Using the last inequality and the definition of $\psi(x)$, we get:

$$\frac{2}{n}f_n \geq \frac{1}{n} \sum_{i=1}^n \psi(\eta_i) \geq \psi\left(\frac{\eta_1 + \eta_2 + \dots + \eta_n}{n}\right) = \psi(0)$$

by Jensen's Inequality. As such, we have illustrated that $\gamma \geq \psi(0)$. But we can find $\psi(0)$ by describing ψ , which was defined as the largest convex function that does not exceed μ and θ . Investigating, we see that μ and θ are convex functions themselves and that $\mu(0) = \theta(0) = 1$. Furthermore, for these functions, $\mu(x) > \theta(x)$ for $x < 0$, and $\mu(x) < \theta(x)$ for $x > 0$. Finding the function ψ means finding the maximum "convex hull," or in this case, a common tangent line $y = mx + b$ for these two functions, connecting $(x_1, 2(e^{\frac{x_1}{2}} + e^{x_1})^{-1})$ and (x_2, e^{-x_2}) , where $x_1 < x_2$. This satisfies the conditions of convexity and maximality of ψ simultaneously, so the function is defined as:

$$\psi(x) = \begin{cases} 2(e^{\frac{x}{2}} + e^x)^{-1} & \text{if } x < x_1 \\ mx + b & \text{if } x_1 \leq x \leq x_2 \\ e^{-x} & \text{if } x > x_2 \end{cases}$$

Finding $mx + b$ ultimately comes down to finding x_1 and x_2 . This can be done by solving the system of equations:

$$\begin{cases} \mu'(x_2) = \theta'(x_1) \\ \mu'(x_2)(x_1 - x_2) + \mu(x_2) = \theta'(x_1) \end{cases}$$

From here, we find that $x_1 = -0.20081\dots$ and $x_2 = 0.155195\dots$. Therefore, $\psi(x) = mx + b$ over $-0.20081\dots \leq x \leq 0.155195\dots$ and $\psi(0) = b$, which can be calculated to be $\gamma = 0.989134\dots$. This is known as Drinfeld's Constant. Therefore, $\frac{2}{n}f_n \geq \gamma \geq \psi(0) \Rightarrow f_n \geq \frac{n}{2} \cdot 0.989134\dots$

(Note: Drinfeld's full method relied on proving the equality $\gamma = \psi(0)$ based on a previous work by Robert Rankin, a Scottish mathematician who examined γ as a sequence for $n \rightarrow \infty$ in a previous paper. He proved that $\lim_{n \rightarrow \infty} \gamma$ converges and that $\lim_{n \rightarrow \infty} \gamma = \inf \gamma$. Since $\inf \gamma$ is really what we are looking for, all that is left is to evaluate $\lim_{n \rightarrow \infty} \gamma$, and Drinfeld proved that $\lim_{n \rightarrow \infty} \gamma = \psi(0)$.)

7 Conclusion

The Shapiro Inequality itself has been resolved through the efforts of many mathematicians over the course of decades. It is worth noting that much of the problem faced by mathematicians originated in the very nature of Drinfeld’s Constant, 0.989134.... This number is very close to 1, which explains why mathematicians had such difficulty in finding counterexamples. *Post facto*, we can reason that since Drinfeld’s Constant provides a strict lower bound, any contradiction that had been found to Shapiro’s Inequality in the past turned out to lie between $\frac{n}{2} \cdot 0.989134\dots$ and $\frac{n}{2}$. And finding the right numbers to fall within that tiny interval is not easy!

References

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Representations of Numbers in Various Bases

Leonna Wang



1 Introduction

Any fraction $\frac{1}{n}$ can be represented in a base b as a repeating or finite decimal. When all prime factors of n are also factors of b , $\frac{1}{n}$ will have a finite decimal expansion. When some of the factors of n are relatively prime to b , $\frac{1}{n}$ will repeat but may have terminating digits at the beginning. Purely periodic decimals occur when $(n, b) = 1$. The numerator need not be 1, but $\frac{1}{n}$ is the most fundamental case. In this article, I will focus on reciprocals of primes to ensure purely periodic decimals.

2 Numerical Data

To begin, I will perform a test where I take a fraction $\frac{1}{p}$ where p is a prime number, and find the period of the repeating digits in its decimal representation. For example, $\frac{1}{11} = 0.090909\dots$ so the period is 2 and the repeating digits are 0 and 9.

p	Period	Repeating Digits
3	1	3
7	6	142857
11	2	09
13	6	076923
17	16	0588235294117647
19	18	052631578947368421
23	22	0434782608695652173913
29	28	0344827586206896551724137931
31	15	032258064516129

37	3	027
41	5	02439
43	21	023255813953488372093
47	46	02127659574468085106382978723404...
53	13	0188679245283
59	58	01694915254237288135593220338983...
61	60	01639344262295081967213114754098...

Define a **reptend prime** as a prime with a cyclic decimal expansion of period $p - 1$. We can notice that the first few reptend primes are 7, 17, 19, 23, 29, 47, 59, 61, 97, 109, 113, 131, 149, 167, 179, 181, 193, ...

3 Period Lengths

Consider the repeating decimal of $\frac{1}{p}$ in base b .

3.0.1 Claim

The period length of $\frac{1}{p}$ is the order of b in U_p .

3.0.2 Proof

First, use the long division algorithm to divide 1 by p in base b . Start by writing 1 as 1.000.... In each step, one must “bring down” a zero and subtract a multiple of p ’s equivalent mod p when dividing by a number b , so long division effectively exponentiates $b \bmod p$. The decimal repeats when the remainder mod p is 1, which occurs when $b^k = 1 \pmod{p}$. Therefore, the period length of $\frac{1}{p}$ is the order of $b \pmod{p}$.

3.1 Lemma

The decimal expansion of $\frac{1}{p}$, where p is a prime number that is not 2 or 5, is purely recurring because the greatest power of 2 and 5 that divides p is 0. More generally, consider a fraction where $\frac{m}{n}$, where m and n are integers and $(m, n) = 1$.

3.1.1 Claim

The number of non-recurring digits is (a, b) , where a is the maximum power of 2 that divides mn and b is the maximum power of 5 that divides mn .

3.1.2 Proof

For any n , we have $n = 2^a \cdot 5^b \cdot m$, where $(m, 10) = 1$ and $m \geq 1$. Now, take the infinite series $\sum_{i \geq 1} a \cdot 10^i \bmod n$. By the Pigeonhole Principle, there must be at least two terms that are congruent mod n . We can set the first such pair of terms to be $a \cdot 10^k$ and $a \cdot 10^{k+l}$, which yields

$$10^k \equiv 10^{k+l} \pmod{n}$$

because the GCD of a and n is 1, as $\frac{a}{n}$ is the simplified fraction. We also have that $(10, m) = 1$, so $1 = 10^l \pmod{n} \Rightarrow 1 = 10^l \pmod{m}$ because $m|n$. Thus, we have

$$a \cdot 10^{k+j} = a \cdot 10^{k+l+m} \pmod{n}$$

for any arbitrary $j \geq 0$. Furthermore, the order l of $10 \pmod{m}$ is the length of the period. Both 2^α and 5^β have to divide a^k as they are relatively prime to $10^l - 1$. The smallest k is therefore the maximum of α and β , representing the length of the non-recurring part of the decimal expansion. Hence, for primes p other than 2 or 5, $\frac{1}{p}$ is purely recurring.

3.1.3 Example

Consider $\frac{m}{n}$ where $m = 3$ and $n = 7$. No factors of 2 or 5 go into 3 or 7, meaning that $a = 0$ and $b = 0$. Therefore, $(a, b) = 0$, and $\frac{3}{7} = 0.\overline{428571}$ has no non-recurring digits.

3.1.4 Example

Consider $\frac{m}{n}$ where $m = 5$ and $n = 8$. Since $5 = 5^1$ and $8 = 2^3$, we have $mn = (5^1)(2^3)$, meaning that $a = 3$ and $b = 1$. We have $(a, b) = 3$, and $\frac{5}{8} = 0.625$ has three non-recurring digits.

4 Period of a Multiple

4.1 Theorem

The period of $\frac{1}{pq}$ is the LCM of the periods of $\frac{1}{p}$ and $\frac{1}{q}$, where p and q are primes greater than or equal to 5.

4.2 Proof

For prime number p , let's set $m = \text{ord}_p(b)$. We have $p \mid -b^m - 1$, so we can also write $b^m = pD + 1$. Therefore,

$$\frac{1}{p}b^m = D + \frac{1}{p},$$

meaning that $\frac{1}{p}$ in base b can be written as a repeating decimal $0.\overline{D}$ with a period of m digits. When m is even, we get

$$pD = b^m - 1 \equiv (-1)^m - 1 = 0 \pmod{b+1} \Rightarrow pD = s(b+1).$$

If we also have $(p, b+1) = 1$, we can further show that $b+1 \mid D$.

Now, we know $1/p$ has period $m = \text{ord}_p(b)$ and $1/q$ has period $n = \text{ord}_q(b)$, and we also have $1/pq$ has period $k = \text{ord}_{pq}(b)$. Finally, we can prove that $k = [m, n]$:

$$\begin{aligned} m = \text{ord}_p(b) \text{ and } p \mid b^k - 1 &\Rightarrow m \mid k \\ n \mid k &\Rightarrow [m, n] \mid k \\ b^m - 1 \mid b^{[m, n]} - 1 &\Rightarrow p \mid b^{[m, n]} - 1 \\ q \mid b^{[m, n]} - 1 &\Rightarrow pq \mid b^{[m, n]} - 1 \Rightarrow k \mid [m, n] \Rightarrow k = [m, n]. \end{aligned}$$

4.3 Example

$\frac{1}{7}$ and $\frac{1}{11}$ have periods 6 and 2, so $\frac{1}{77}$ has period 6 because $[6,2] = 6$.

4.4 Example

$\frac{1}{13}$ and $\frac{1}{17}$ have periods 6 and 16, so $\frac{1}{221}$ has period 6 because $[6,16] = 48$.

5 Half Period Theorem

5.1 Theorem

For primes $p \neq 2, 5$, if $\text{ord}_p(b) = 2d$ for some $d \geq 1$, then

$$\frac{n}{p} = 0.\overline{a_1 a_2 \dots a_d a_{d+1} \dots a_{2d}},$$

where $a_k + a_{d+k} = b$ for $1 \leq k \leq d$.

5.2 Proof

We have the following:

$$\text{ord}_p(b) = 2d \Rightarrow b^{2d} = mp + 1 \Rightarrow \frac{n}{p} \times b^{2d} = mn + \frac{n}{p}$$

$$\frac{n}{p} = 0.\overline{D}, D = \frac{n}{p}(10^{2d} - 1) < b^{2d-1} \Rightarrow D = a_1 a_2 \dots a_d a_{d+1} \dots a_{2d}.$$

Let $A = a_1 a_2 a_3 \dots a_d$ and $B = a_{d+1} a_{d+2} a_{d+3} \dots a_{2d}$, so we have $D = b^d A + B$. Note that both A and B have length d . We also have $p \nmid b^d - 1$ because $2d$ is the order of b for $(\text{mod } p)$. Then,

$$pD = n(b^{2d} - 1) = n(b^d - 1)(b^d + 1) \Rightarrow p \mid n(b^d + 1) \text{ and } b^d + 1 \mid D$$

$$D = b^d A + B \equiv A + B \pmod{b^d - 1} \Rightarrow A + B = b^d - 1 \text{ because } A + B < 2(b^d - 1).$$

5.3 Example

The decimal expansion of $\frac{1}{7}$ is $7 \ 0.\overline{142857}$, and $142 + 857 = 999$.

5.4 Example

The decimal expansion of $\frac{1}{29}$ is $0.\overline{0344827586206896551724137931}$, and $142 + 857 = 999$. $03448275862068 + 96551724137931 = 99999999999999$.

6 Connections to Class Numbers

6.1 Conjecture

If $\frac{1}{p}$ has a cyclical permutation, then if the alternating sum of the digits in the decimal expansion is taken: $\overline{a_1 a_2 \dots a_d}$ then $\sum_{i=1}^d (-1)^i \cdot a_i$ is equal to 0 when the prime is of the form $4k + 1$ and $11h(-p)$ when the prime is of the form $4k + 3$. $h(-p)$ is equal to the class number of the primitive binary quadratic forms with discriminant $-p$.

6.2 Motivation

Below are a few values:

p	Sum	$h(-p)$
7	11	1
17	0	4
19	11	1
23	33	3
29	0	6
47	55	5
59	33	3
61	0	6
97	0	4

Fields looked into: Stark-Heegner Theorem, Imaginary Quadratic Fields, Ideals...

I explored different sums of $\frac{1}{p}$ which yielded the formula in Section 4, and eventually took a look at the alternating sums. I then ran a Python script for the numbers this sum yields for $\frac{1}{p}$. Some of the data displayed that some of the numbers with this sum as multiples of 11 are primes with period $p - 1$. This led to the splitting into cases of primes of the form $4k + 1$ and $4k + 3$. Primes of the form $4k + 1$ and $4k + 3$ had a sum of 11 multiplied by some integer.

6.3 Example

For the decimal expansion of $\frac{1}{23}$, this sum equals $33 = 3 \cdot 11$, where $3 = h(-23)$, the class number of -23 . The forms of the binary quadratics are $(a, b, c) = (1, 1, 6), (2, -1, 3), (2, 1, 3)$. 23 is additionally of the form $4k + 3$ where $k = 5$.

6.4 Example

For the decimal expansion of $\frac{1}{17}$, this sum equals 0. 17 is of the form $4k + 1$ where $k = 4$.

7 p -adic Integers and Decimal Expansions

7.1 Theorem

Let $f(x) \in \mathbb{Z}[x]$, $a, p \in \mathbb{Z}$, p prime, and suppose $f(a) \equiv 0 \pmod{p^n}$ and $f'(a) \not\equiv 0 \pmod{p}$. Then there is a unique $a_n \in \mathbb{Z}_{p^{n+1}}$ such that $f(a + a_n p^n) \equiv 0 \pmod{p^{n+1}}$.

7.2 Proof

We can set $m = \deg(f(x))$ and expand $f(x)$ as a Taylor Series at $x = a$ to get:

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \dots + \frac{f^{(m)}(a)}{n!}(x - a)^m.$$

Next, evaluating at $x = a + a_n p^n$ yields

$$f(a + a_n p^n) = f(a) + f'(a)a_n p^n + \dots + \frac{f^{(m)}(a)}{n!}a_n^m p^{nm}.$$

Since we are working $(\text{mod } p^{n+1})$, we can reduce

$$f(a + a_n p^n) \equiv f(a) + f'(a) a_n p^n \pmod{p^{n+1}}$$

as all other terms have higher than p^{n+1} powers of p and have integer coefficients as you can express $\frac{1}{m!} f^{(m)}(x) = \binom{n}{m} x^{n-m}$.

We want to prove that there exists an a_n such that $f'(a) a_n p^n \equiv -f(a) \pmod{p^{n+1}}$. Because we have $p^n | f(a)$, we can use modular cancellation to get that

$$f'(a) a_n p^n \equiv -f(a) \pmod{p^{n+1}} \iff f'(a) a_n \equiv \frac{-f(a)}{p^n} \pmod{p}.$$

Using linear diophantine equations and the fact that $f'(a) \not\equiv 0 \pmod{p}$, there is a unique solution for $a_n \pmod{p}$. Thus, if we trace backwards, proving this means proving that $f(a + a_n p^n) \equiv 0 \pmod{p^{n+1}}$, thus completing the proof.

This can also be generalized to work over p -adic integers.

8 Polynomials

We can extend the ideas from considering fractions in base b to considering the fractions in various polynomial rings. However, first note the following remark.

8.0.1 Remark

A number in base b , such as $(a_1 a_2 a_3 \dots a_k)_b$ can simply be written as a polynomial in b . So, we can write

$$(a_1 a_2 a_3 \dots a_k)_b = a_1 b^{k-1} + a_2 b^{k-2} + a_3 b^{k-3} \dots + a_k.$$

Likewise, if we have a polynomial in b , call it $f(b)$, then we can view $f(b)$ as a number in base b by examining the coefficients of f .

For example, suppose we have the polynomials $\text{mod } p : \mathbb{Z}_p$. Now, consider the fractions $\frac{1}{x+u}$ of $\mathbb{Z}_p$, where u is a unit in $\mathbb{Z}_p$. Depending on the values of p and u , the polynomial expansion of $\frac{1}{x+u}$ in base p can either be infinite or finite.

8.0.2 Example

Suppose $p = 5$, so we are working in $\mathbb{Z}_5$. Consider the selection $u = 1$. Then we have that

$$\begin{aligned} \frac{1}{x+1} &= \frac{1}{1-(-x)} \\ &= 1 - x + x^2 - x^3 \dots \\ &= 1 + 4x + x^2 + 4x^3 \dots \text{ in } \mathbb{Z}_5, \end{aligned}$$

which is an infinite polynomial.

Note that this infinite polynomial only has coefficients 1, 4, which is not the entire set of units in $\mathbb{Z}_5$. Specifically, the polynomial does not have coefficients 2 or 3. This begs the question, when do the coefficients of the polynomial expansion of $\frac{1}{x+u}$ in $\mathbb{Z}_p$ cycle through all the elements of U_5 ? The following conjecture can thus be made:

8.1 Theorem (Generators Conjecture)

When written as a polynomial in $\mathbb{Z}_p$, the fraction $\frac{1}{x+u}$ will only cycle through all the elements of U_5 if $u = g$, where g is a generator mod p .

8.1.1 Example

As shown previously, since $u = 1$ is not a generator mod 5, the expansion of $\frac{1}{x+1}$ does not cycle through all coefficients. However, now suppose $u = 2$, which is a generator mod 5. This means that the polynomial expansion in $\mathbb{Z}_5$ is

$$\begin{aligned}\frac{1}{x+2} &= \frac{\frac{1}{2}}{\frac{x}{2}+1} \\ &= \frac{\frac{1}{2}}{\frac{x}{2}+1} \\ &= \frac{\frac{1}{2}}{1 - (-\frac{x}{2})} \\ &= \frac{3}{1 - (-3x)} \\ &= 3 - 9x + 27x^2 - 81x^3 + 243x^4 \dots \\ &= 3 + x + 2x^2 + 4x^3 \dots \text{ in } \mathbb{Z}_5\end{aligned}$$

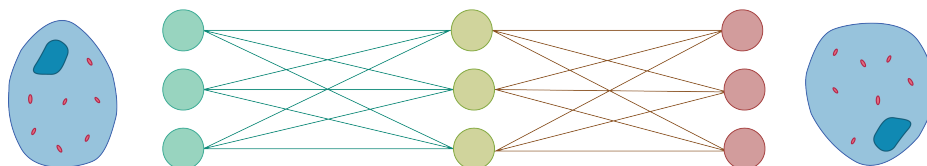
which certainly hits all elements of U_5 .

9 Conclusion

When diving into the world of the representation of numbers, one can figure that there are many conjectures that are yet to be made. Although a lot of information can currently be found online, sometimes it is nice to relax and attempt to make conjectures and prove or salvage them on your own. The conjectures that I made on this topic range from periods to relations to calculus (Taylor function), and there are still many more phenomena that are to be desired in this field of mathematics.

Graph Representation Learning in Cell Biology

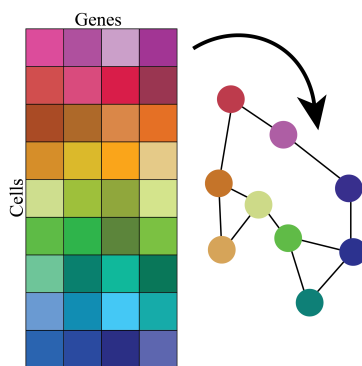
Galen Jack



1 Single-Cell RNA Sequencing Data

Though mathematics and biology have long been disparate fields, computer scientists, mathematicians and biologists are beginning to collaborate. They are taking advantage of the recent technological advancements which have allowed biologists to collect cell data on a single-cell scale, as well as advances in machine learning and mathematics which allow computers to process these massive data sets. To collect single-cell data, biologists use RNA sequencing, which enables scientists to study the set of all messenger RNA (mRNA) molecules in a cell also known as the transcriptome. This technology can be used to compare individual cell transcriptomes, so scientists can discern the heterogeneity between cells, which is immensely useful in cancer research. Additionally, single-cell RNA sequencing data provides information on gene expression at the single-cell level.

From this data, a matrix can be constructed in which rows refer to individual cells, while columns refer to genes. A k -nearest neighbors graph can be constructed by giving a vertex on the graph the labels of its k nearest neighbors. In this case, the nearest neighbors are found by a genetic similarity metric, resulting in the connection of cells that are genetically similar.



2 Graph Representation Learning

Neural networks are networks of nodes or artificial neurons that each have a threshold. When the input these nodes receive exceeds their threshold value, they return an output. Each node also has a weight which determines how strong its output is. After the network of nodes receives an input, it sums its outputs and passes it to the next layer or network of nodes. These networks train on labeled sets, comparing their output to the label of the input to determine the optimal weights for their nodes. Using the vast data generated by single-cell RNA sequencing as a training set for neural networks allows these networks to make predictions on biological factors that can help predict cancer types.

To achieve this functionality, nodes on the graph are mapped to vector space, also referred to as latent space. Graph representation learning can occur in either an unsupervised, supervised, or semi-supervised setting. In an unsupervised setting, the most relevant property of the graph for the question at hand must be identified. In this way, unsupervised learning analyzes and clusters a data set and discovers patterns without manual help. Supervised learning, on the other hand, uses labeled data sets and can sort data into categories or analyze the relationships between dependent and independent variables. Semi-supervised learning combines these two, using a small set of labeled training data and a larger set of unlabeled data. In the context of RNA-sequencing data, unsupervised learning can create cellular similarity graphs without human intervention, while semi-supervised learning can classify cancers by using a labeled set of cancerous cells.

Graph neural networks accept a graph as an input and pass a graph as the output, updating the information in the nodes and edges without changing the actual connectivity of the input graph. Because of how broad graphs are, it is difficult to represent them in a format that computers can accept. To tackle this problem, the success of convolutional neural networks can be generalized from grid-like structures to arbitrary graphs. This generalization can be achieved through polynomial filters on graphs.

Before we proceed, let's define some terms:

- An **adjacency matrix** is a square matrix that can be used to represent a graph. The elements of the matrix indicate the adjacency of the vertices on the graph. For any two vertices on the graph, there exists an entry in the matrix that indicates if there exists an edge between the vertices (1 for an edge, 0 for no edge).
- For some graph G with adjacent matrix A , the **diagonal degree matrix** D can be constructed with:

$$D_{vu} = \sum_u A_{vu}$$

where A_{vu} denotes the entry in row v and column u of adjacency matrix A .

- The **Graph Laplacian** L is a square matrix defined as $L = D - A$. From the graph Laplacian, polynomials of the form:

$$w_0 I_n + w_1 L + w_2 L^2 + \dots + w_d L^d$$

can be constructed. These polynomials act as filters, with the coefficients behaving as the weights of the filters. To perform a convolution on a graph, nodes can be stacked to create a vector x . A convolution x' upon the graph with vector x can be expressed as

$$x' = p_w(L)x$$

This means that convolutions at node v occur only with nodes u that are no more than d steps away. The attributes of a node are combined with those of its immediate neighbors through this process, based on the coefficients and degree of the polynomial. To construct a graph neural network from this idea of convolving by a polynomial filter, the coefficients or weights of the filter are set as learnable parameters for the neural network.

3 Applications

Graph convolutional networks are powerful tools capable of deciphering large datasets. An example of this capability is a study which analyzed 913 cells from the cortex of a mouse, including thousands of interactions between ligands and receptors. Ligands are molecules or atoms that bind to a protein, transmitting a signal. Graph convolutional networks were shown to be 20% more effective than previous methods in detecting receptor-ligand interactions from spatial data.

Additionally, graph convolutional networks can be used to predict the directionality of other interactions. Using ligand-receptor interactions whose directionality is known (the ligand interacts with the receptor, not the other way around), graph convolutional networks were able to discern the directionality of other pairs.

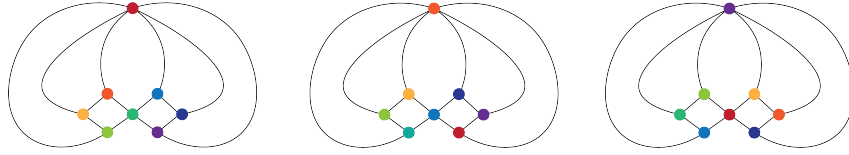
Finally, graph convolutional networks have also been used to predict the function of genes. Proteins have a similar function as proteins they interact with, leading to the “guilt by association” method of predicting protein function. Graph neural networks serve as an alternative, using spatial data to predict protein associations. Since Genes are composed of proteins, both of these methods can be applied to genes as well. Convolutional graph networks outperform “guilt by association” predictions by a large margin. Being able to accurately predict the function of genes or of biological interactions is important because they allow scientists and doctors to better treat patients and develop medications.

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Matroid Intersection: Generalizing Problems in Combinatorial Optimization

Shuichi Kinoshita



1 Introduction

Matroids are an abstraction of the concept of linear independence from linear algebra. There are many applications of matroids, but one particularly famous application is the matroid intersection. Below are interesting example problems with matroids.

1.1 Colorful Weights

Given a set of elements, each with some weight and unique color, find the maximum weight that can be obtained by selecting a subset of elements such that no element in the subset has the same color.

1.2 Colorful Graph

Given a graph with each edge assigned a color, find a minimum spanning tree where all edges are distinct.

2 Definitions

A **matroid** $M = (S, \mathcal{I})$ is a set S with a collection of subsets $\mathcal{I}$ such that the axioms below hold:

1. For $J \in \mathcal{I}$ and $I \subseteq J$ then $I \in \mathcal{I}$.
2. Given $I, J \in \mathcal{I}$ with $|I| > |J|$, there exists element $x \in I \setminus J$ such that $J \cup \{x\} \in \mathcal{I}$.
3. The empty set $\emptyset$ is in $\mathcal{I}$.

2.1 Terminology

A set I is called **independent** if $I \in \mathcal{I}$ and otherwise **dependent**.

The set S in $M = (S, \mathcal{I})$ is called the ground set of M .

A **basis** is an independent set with maximal size. Notice that every basis has the same size due to Axiom 2, above. If there exists a bigger independent set, one can include the element from it to achieve the same size. Knowing the basis can completely characterize the matroid.

A **circuit** is a dependent set which becomes independent if any element is removed. Note that given two circuits in a matroid that share an element but are not equal, a circuit with elements from both that does not include the shared element can be constructed. Circuits can also characterize a matroid.

An **independence oracle** is a function which determines whether a subset is independent or not in a given matroid. This is used to calculate the time complexity of algorithms involving matroids instead of the regular asymptomatic notation, because the time taken to test independence depends on the matroid. However, it still is useful to characterize how fast an algorithm runs.

A **rank** of a matrix is a function r such that $r(A)$ for some $A \subseteq S$ gives the size of maximum independent subset of A . There are a few properties that will be useful later:

- The rank of a set cannot be greater than its size.
- The rank of an independent set is its size.
- The rank of any subset of S would not be greater than $r(S)$.
- Including an element increases the rank by 1 or keeps it the same. Formally, $r(S) \leq r(S \cup \{x\}) \leq r(S) + 1$.
- Finally, $r(A) + r(B) \geq r(A \cup B) + r(A \cap B)$. Intuitively, this means that it is not always better to put all your eggs in one basket.

2.2 Examples

Let's start with the simplest possible matroid: a uniform matroid. This consists of a ground set S , and any subset $A \subseteq S$ is considered independent if $|A| \leq k$ for some k . The trivial matroid is the case when $k = 0$, that is no set beside the empty set is independent. The complete matroid is when $k = |S|$, that is every subset of S is independent. In this matroid, the basis would be any subset of size k and the circuit would be any subset of size $k + 1$. The rank function would simply be $r(S) = \max(|S|, k)$.

Another simple example is the colorful matroid. It consists of elements, each of which has a color, not necessarily different from the colors of other elements associated with it. A subset is considered independent if it does not contain elements of the same color. We see this follows Axiom 1, because a set with distinct colors will still have distinct colors if you take some subset of them. It follows Axiom 2 because given $|I| > |J|$ there must be some color that we have not used in I . Here, a basis would contain all the distinct colors available in the matroid. The circuit is

more interesting and will consist of two elements of the same color. (Note that it is really easy to see here why a dependent set must contain a circuit.) The rank function would simply count the number of distinct colors in the matroid.

A graphic matroid has a ground set that is a set of edges in some graph, and a subset that is considered independent if it does not contain a cycle. It follows Axiom 1 as it is impossible to introduce a cycle by removing edges. Proving that Axiom 2 is satisfied is harder, but it should be intuitive upon trial and error. The basis of the graphic matroid is set of $n - 1$ edges which connect all n vertices of the graph, called the minimal spanning tree. The circuit would be any cycle in the graph. The rank function becomes very simple as it only returns one less than the number of vertices in the graph (that is, if the graph is connected).

A linear matroid has a ground set of vectors in a vector space, and they are all independent if they are linearly independent. This is where the theory of a matroid originates from, along with the graphic matroid. Any basis of the vector space and the rank of the matroid also corresponds to the rank of the vector space. The circuit does not have any analogue in linear algebra, however.

A truncated matroid can be obtained from any matroid by simply restricting the maximal size of an independent set to be k . For example, we could take a colorful matroid with n colors and truncate it by taking the maximal independent set to be k . Then, any set would be independent if it doesn't contain any duplicate colors and contains at most k distinct colors. Intuitively, this is possible because the subset property of independence does not break when taking the maximum, and the second property of picking a greater independent set becomes infeasible if one didn't exist.

3 Rado-Edmond Algorithm

Consider the problem of finding an independent set with maximal weight. We assign a weight $w : S \rightarrow \mathbb{R}$ to each of the elements in our ground set and $w(I) = \sum_{e \in I} w(e)$. We seek to find $\max_{I \in \mathcal{I}} w(I)$. This is Problem 1.1, Colorful Weights, at the beginning of this article.

3.0.1 Claim

The maximum can be achieved by a greedy algorithm as follows:

1. Sort each of element e in S in descending order of $w(e)$.
2. Start with $I = \emptyset$ and include e in I if $I \cup \{e\}$ is independent.

We see that it takes $\mathcal{O}(n)$ oracle calls and $\mathcal{O}(n \log n)$ operations.

3.0.2 Proof

We will prove this by induction. We see that $\emptyset$ is optimal for size 0. Let A be optimal for independent set size n and B be optimal for $n + 1$. Then, by Axiom 2, we have an element $x \in B \setminus A$ such that $A \cup \{x\}$ is independent. Since A and B are both optimal,

$$\begin{aligned} w(A \cup \{x\}) &\leq w(B) \\ w(B \setminus \{x\}) &\leq w(A) \\ w(B) &\leq w(A \cup \{x\}) \\ w(B) &\leq w(A \cup \{x\}) \leq w(B) \end{aligned}$$

Therefore $w(B) = w(A \cup \{x\})$. Thus, for $w(B)$ to be minimal we need $w(x)$ to be minimal.

4 Matroid Intersection

The problem of matroid intersection is given two matroids $M_1 = (S, \mathcal{I}_1)$ and $M_2 = (S, \mathcal{I}_2)$ with the same ground set S , we wish to find a set I with maximum size such that $I \in \mathcal{I}_1 \cap \mathcal{I}_2$. Given this, we see that Problem 1.2, Colorful Graph, reduces to finding the intersection of a colorful matroid and a graphic matroid. Other examples include maximal bipartite matching (where we take each of the matroid to be the sets independent if the set of edge doesn't share a vertex on one side).

4.1 Strong Basis Exchange

To find an algorithm that performs this is quite a challenge. Therefore, we must first make some observations. The **Strong Basis Exchange** states that given two bases A and B , it is possible to pick elements $x \in A \setminus B$ and $y \in B \setminus A$ such that $A \setminus \{x\} \cup \{y\}$ and $B \setminus \{y\} \cup \{x\}$ are both bases. This can be extended to any independent sets $|I| = |J|$ since they can be treated as a basis in the truncated matroid.

Before we prove this, we must build some intuition on how that might work. We see that in a colorful matroid, we can do this by either picking a color that is in $I \setminus J$ and a color that is in $J \setminus I$ or, if it is not possible, then all the colors are the same, so we could swap two elements that have the same color.

To do this we define $\text{span}(A) = \{x \in S \mid r(A) = r(A \cup \{x\})\}$. This is analogous to the definition of span in linear algebra. Now intuitively,

- If $A \subseteq B$ then $\text{span}(A) \subseteq \text{span}(B)$
- If $e \in \text{span}(A)$ then $\text{span}(A) = \text{span}(A \cup \{e\})$

These can obviously be proven rigorously but should be intuitive, although how to prove them might not be obvious.

Using these, we prove the Strong Basis Exchange lemma, which is the following:

4.1.1 Claim

Given two bases A and B , it is possible to pick elements $x \in A \setminus B$ and $y \in B \setminus A$ such that $A \setminus \{x\} \cup \{y\}$ and $B \setminus \{y\} \cup \{x\}$ are both bases.

4.1.2 Proof

Suppose $x \in A \setminus B$. Since B is a basis, $B \cup \{x\}$ must contain a unique circuit, call it C . Now this circuit C must contain x since all the subset of B is independent.

Since C is a circuit, we have $r(C) = r(C \setminus \{x\})$. This further implies that, because

$$C \setminus \{x\} \subseteq (A \cup C) \setminus \{x\}$$

and therefore

$$\text{span}(C \setminus \{x\}) \subseteq \text{span}((A \cup C) \setminus \{x\}),$$

we have $x \in \text{span}(C \setminus \{x\})$ because $r(C) = r(C \setminus \{x\})$. Then, we can conclude that $x \in \text{span}(C \setminus \{x\})$. Now, we have that

$$\text{span}((A \cup C) \setminus \{x\}) = \text{span}(A \cup C) = S,$$

where the last step follows since A is a basis; that is, adding any element does not change its rank.

Then $\text{span}((A \cup C) \setminus \{x\})$ must contain a basis, call it X ; otherwise it cannot span S as it was shown to be. Then $A \setminus \{x\}$ is independent, as any subset of an independent set is, and X is independent with $|X| > |A|$, since all basis have the same size. So, there exists $y \in X \setminus (A \setminus \{x\})$ such that $A \setminus \{x\} \cap \{y\}$ is independent. But we have

$$X \setminus (A \setminus \{x\}) \subseteq \text{span}((A \cup C) \setminus \{x\}) \setminus (A \setminus \{x\}) \subseteq C \setminus \{x\},$$

so in fact y must also be in $C \setminus \{x\}$. Now recall that C was a circuit within $B \cup \{x\}$ so $C \setminus \{x\} \subseteq B$. Since C was the unique circuit in $B \cup \{x\}$, $B \setminus \{y\} \cup \{x\}$ must also be independent because removing y breaks the circuit introduced by adding $\{x\}$.

4.2 Main Algorithm

For a matroid $M = (S, \mathcal{I})$ and $I \in \mathcal{I}$, the exchange graph $D_M(I)$ is defined as a **bipartite graph** between I and $S \setminus I$ with edges between $x \in I$ and $y \in S \setminus I$ if $I \setminus \{x\} \cup \{y\}$ is independent.

Now if I and J are both independent with $|I| = |J|$, then there exists a matching between $I \setminus J$ and $J \setminus I$ in $D_M(I)$. By the Strong Basis Exchange, we can find an edge in $D_M(I)$ between $I \setminus J$ and $J \setminus I$ and updated I to $I' = I \setminus \{x\} \cup \{y\}$ and $J' = J \setminus \{y\} \cup \{x\}$. This can continue until the sets are the same.

However, the converse is not true, and it is easy to come up with a counter example. To fix this, we need to add a condition specifying that matching must be unique. This happens to be a necessary and sufficient condition for an independent set.

We can also combine the exchange graph for both M_1 and M_2 . The directed bipartite graph $D_{M_1, M_2}(I)$ for $I \in \mathcal{I}_1 \cap \mathcal{I}_2$ has bipartitions I and $S \setminus I$ and has edge from I to $S \setminus I$ when $I \setminus \{y\} \cup \{x\} \in \mathcal{I}_1$ and edge from $S \setminus I$ to I when $I \setminus \{x\} \cup \{y\} \in \mathcal{I}_2$.

Now we can finally state the algorithm to find the intersection:

1. Set $I = \emptyset$.
2. Find $X_1 = \{x \notin I \mid I \cup \{x\} \in \mathcal{I}_1\}$ and $X_2 = \{x \notin I \mid I \cup \{x\} \in \mathcal{I}_2\}$.
3. Find an augmenting path in $D_{M_1, M_2}(I)$, a path from an element in X_1 to an element in X_2 which does not contain shortcuts (edge that point from an earlier vertex on the path to a non-adjacent later vertex on the path).
4. If there is no such path, I is the maximal intersection.
5. Otherwise, we replace I with $(I \setminus P) \cup (P \setminus I)$ and go back to Step 2.

4.2.1 Claim

The output I is independent in both $\mathcal{I}_1$ and $\mathcal{I}_2$.

4.2.2 Proof

We can assume that at the start I is an independent set in both M_1 and M_2 . Then, we want to prove $(I \setminus P) \cup (P \setminus I)$ is also an independent set in both M_1 and M_2 .

First, we show that $I \in \mathcal{I}_1$. We define $M'_1 = (S \cup t, \{J \mid J \setminus \{t\} \in \mathcal{I}_1\})$ or in standard terms, M_1 with added element t which is independent to all the elements in M_1 . We can define M'_2 analogously. Then, we know the path P combined with t creates a matching between $I \cap P \cup \{t\}$ and $S \setminus I$. This matching is unique since P does not contain any shortcut. Therefore, $(I \setminus P) \cup (P \setminus I)$ is independent in $\mathcal{I}_1$.

Similarly, I is independent in $\mathcal{I}_2$.

4.2.3 Claim

The output I is largest I possible.

4.2.4 Proof

To prove this, we will show that

$$\max_{I \in \mathcal{I}_1 \cup \mathcal{I}_2} |I| = \min_{U \subseteq S} r_1(U) + r_2(S \setminus U).$$

We see that for any $U \subseteq S$ and $I \in \mathcal{I}_1 \cup \mathcal{I}_2$,

$$\begin{aligned} |I| &= |I \cap U| + |I \cap (S \setminus U)| \\ &\leq r_1(U) + r_2(S \setminus U) \end{aligned}$$

since $I \cap U \in \mathcal{I}_1$ and $I \cap (S \setminus U) \in \mathcal{I}_2$.

If we could come up with $U \subseteq S$ and $I \in \mathcal{I}_1 \cup \mathcal{I}_2$ such that the equality was satisfied, then the maximal size of I would be $|I|$ since anything greater will violate the condition that $|I|$ should be less than the minimum with a given U .

If we set $U = \{z \in S \mid z \text{ has a path to a vertex in } X_2\}$, then we claim that $r_1(U) = |I \cap U|$ and $r_2(S \setminus U) = |I \cap (S \setminus U)|$.

If $r_1(U) \neq |I \cap U|$, then since $I \cap U \subseteq U$ and $I \cap U \in \mathcal{I}_1$, we can see that $r_1(U) > |I \cap U|$. Then by the Axiom 2, there exists $x \in U \setminus I$ such that $(I \cap U) \cup \{x\} \in \mathcal{I}_1$. Then we have that either $I \cap \{x\}$ is independent or contains a unique cycle C which contains at least one element not from $(I \cap U) \cup \{x\}$. Either way, we have $y \in I \setminus U$ such that $I \setminus \{y\} \cap \{x\} \in \mathcal{I}_1$ so there exists a directed edge from y to x . Then $y \in U$, since $x \in U$, which is a contradiction.

We can similarly prove that $r_2(S \setminus U) = |I \cap (S \setminus U)|$.

5 Conclusion

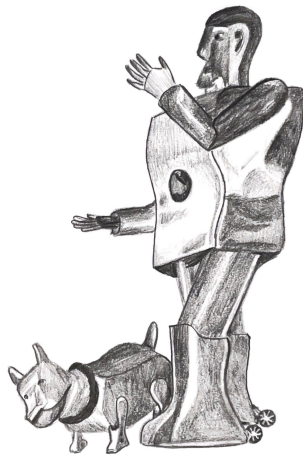
There are countless applications for matroids. Matroids are fundamental in the study of combinatorial optimization, and some other practical techniques include matroid unions and matroid partitions. Furthermore, there are other areas of research in matroids, such as whether a matroid can be represented as a matrix.

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Will Robots Take Over the World? The Effects of AI on Jobs

Aditya Anand



Sparko + Electro

1 Introduction

When you think of the future, you may envision a world run by robots, where humans can relax and sit back while an android does your chores for you. Studies show that this may become reality sooner than you may expect.

2 The Past: The History of Robots and AI

2.1 The Beginning

Artificial Intelligence is defined as “the ability of a computer or a robot controlled by a computer to do tasks that are usually done by humans.” The idea of robots was first presented in 1920 by Karel Capek, in his play *Rossum’s Universal Robots*. The play presented robots taking over the world, giving birth to the science fiction genre.

The first robots were displayed in 1939 at the World’s Fair by the Westinghouse Electric Company. “Elektro the Moto-Man” had the ability to walk, talk, smoke cigarettes, blow up balloons, and move its head and arms. Elektro appeared again in the 1940 World’s Fair, along with Sparko, a robot dog who could sit, bark, and beg.

Following the revolutionary creations of Elektro and Sparko, Isaac Asimov, a professor at Boston University, created the three laws of robotics:

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey orders given to it by human beings except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

These laws showed that robotics was beginning to be recognized as a prominent field of science.

2.2 Artificial Intelligence



Figure 1: W. Gray Walter and one of his robots

While Elektro and Sparko were advanced for their time, they required voice commands to function. The first official artificial intelligence came in the form of two robot tortoises, Elmer and Elsie. Built by William Grey Walter in the late 1940s, these autonomous robots utilized sensory inputs by perceiving light and feeling with a bump sensor. These advancements developed not only technology but also our understanding of the human nervous system.

3 The Present: Modern Advancements

3.1 Horses

Many people question whether robots can take over humans' jobs. However, machines have already made one work animal obsolete: horses.

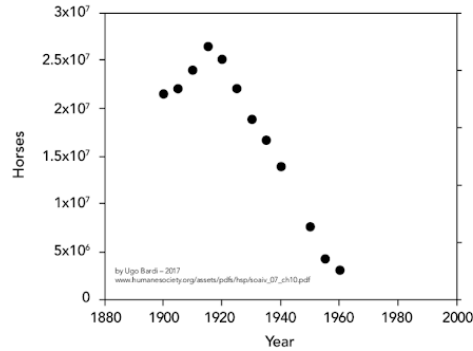


Figure 2: Horse population over time

As the graph depicts, the population of horses peaked in 1915 but sharply declined soon after. This is directly linked to the fact that cars began to be commonly used in the 1920s. As cars became more available, horses became unnecessary for human use. Other roles traditionally held by horses were also impacted. Technology started to replace horses in transportation, agriculture, warfare, and more.

3.2 Current Robots/AI Replacing Jobs



While there may be doubts about how quickly robotic replacement can happen with humans, there are already intelligent robots among us. An example of this is Baxter, a robot with vision who can “learn what you want him to do by watching you do it.” Baxter’s skill set includes industrial tasks such as “loading, unloading, sorting, and handling of materials.” While the robot takes much more time performing tasks when compared to a human, it is also much more cost-effective. Baxter costs about \$25,000 to create and maintain, which is about \$40,000 less than the average household income in America.

While this may sound like a complex idea, there are already examples of artificial intelligence replacing human jobs. In new supermarkets, robotic cashiers have replaced humans; “what used to be 30 humans is now 1 human overseeing 30 cashier robots.” Another example is the surge in robot baristas; over 500,000 Americans work as baristas, and their jobs are in danger of being replaced by artificial intelligence.

Many other fields are also impacted by robots. Chatbots now lead customer service and can help users do anything, from opening an account to transferring money. In addition, self-driving cars are emerging. The transportation industry currently consists of at least 70 million workers worldwide. Self-driving cars will likely replace these human drivers just as cars replaced horses. Ultimately, robots' ability to think quickly with low chance of error allow them to take human jobs.

3.3 Professional Bots

Careers such as lawyers and doctors require years of education and hours of dedicated work every day. However, lawyers and doctors are still humans, and in the words of Alexander Pope, “to err is human.” For example, lawyers must look through countless legal documents to find details supporting their case. Missing just one paragraph can severely increase their workload. Professional bots can look through millions of emails or documents in a matter of hours, whereas it would take humans weeks. The bots outweigh humans in cost efficiency, time, and accuracy. All of these factors lead to a boost in the economy as well, since more work is accomplished in the same amount of time.

Artificial intelligence can also leave a mark on the medical industry. Humans are not capable enough to understand all medicines and their interactions with each other. By analyzing symptoms and comparing them to a database, robots can accurately diagnose patients and prescribe them the necessary drugs. While it is impossible for a human to know how every medicine affects the human body, this is not an issue for robots. Bots can also use the experience of other robots to improve themselves.

Watson, a robot created by IBM, is providing guidance for lung cancer treatments at Memorial Sloan Kettering Medical Center. Watson is an example of how artificial intelligence can perform human tasks more efficiently. Watson has also made a guest appearance on ‘Jeopardy!’, where it showcased its ability to quickly solve a variety of trivia questions.



While lawyers and doctors will most likely stay, their need and numbers will decrease in the near future.

3.4 Self-Teaching Bots

Breakthroughs in technology have even reached self-teaching robots. This may sound preposterous, but these bots are essential to America's economy. The stock market, which used to be run by humans, is now run by “bots who taught themselves to trade stocks trading stocks” with other

such bots. This is a matter of concern to human workers, as it shows the capability of artificial intelligence to learn and replace most, if not all, jobs.

4 The Future: Paradise or Dystopia?

4.1 Long Term Effects of AI on Jobs

Robots already play a significant role in our economy. As they begin to take more jobs from people, humanity will be drastically impacted. It's not wrong for people to believe that technology will make life easier and allow more leisure for humans. However, convenience comes with a cost. In the same way that technology took away horse labor, AI can have detrimental effects on human labor.

There are many debates about whether or not AI will replace all jobs. Some argue that bots can teach themselves to do any work, while others say that careers which involve emotions, such as therapists, are safe. Nevertheless, most agree that robots can and will replace most jobs. Since AI is more efficient than humans, the economy will likely grow, while the income of humans will decrease.

4.2 A Proposed Solution

The coming wave of AI is undeniable. The only action humanity can take is to prepare for it. Some politicians agree that the best way to help the citizens is a Universal Basic Income, or UBI, where the government pays a set amount of money to all citizens. This might help families sustain themselves even when their job is replaced by AI. Others are opposed to this idea, believing that it will make people underproductive and overdependent on the government.

5 Conclusion

Robots will continue to replace humans in the work world; it is not a matter of if, but rather, when. Politicians and scientists must work together and find a way to help people with diminished income or obsolete jobs. If they don't, people may find their own creations working against them.



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Stuyvesant Faculty Perspectives:
Mr. JonAlf Dyrland-Weaver, Class of 2001:
Computer Science Principles, in Class and in Daily Life

Craig Chen



Mr. JonAlf Dyrland-Weaver has inspired students in computer science for 18 years. After graduating from Stuyvesant in 2001, he majored in Philosophy and minored in Computer Science at Columbia College. He then earned a master's degree in Mathematics Education from the City College of New York. In 2005, he began his career as a teacher at Brandeis High School through the New York City Teaching Fellows Program. At Stuyvesant, he has taught a wide range of courses including Intro to Computer Science, AP Computer Science, Software Development, Systems Level Programming, and Computer Graphics, and has expanded Stuyvesant's computer science department to include courses such as NeXTCS. Mr. DW, as he is popularly known, currently serves as Stuy's Computer Science Coordinator.

Mr. Dyrland-Weaver's answers are taken from an interview conducted by Craig Chen and have been edited for clarity.

1) Why do you enjoy computer science and what makes teaching it meaningful to you?

Computer science is, to me, about solving problems and looking at problems in interesting ways. How do I get information that is useful to people in an accessible and understandable way? Or how do I make a fun game? These are practical problems, but there are also theoretical problems that have to do with questions of computability and complexity. Computer science runs the gamut of problem solving, and in my own eyes, that's what makes the field so rich.

When I teach computer science, I want to portray the learning of the subject as building up a toolbox of techniques, programs, and environments, as well as the understanding of when to

use them. I hope to empower others to develop and use such tools to do other things—to do interesting things. At the end of the day, I am giving my students the ability to solve all sorts of different kinds of problems with computers, which are incredibly useful and powerful tools. The better that my students understand how to use them, the better they're going to be able to do whatever it is, whether it's write software, produce webpages, or do anything else. And so that's what it is for me, and that's why I enjoy teaching. It provides students with freedom of exploration.

2) What do you believe is one of the most underappreciated aspects of being a computer science teacher?

It would be the amount of time and thought that goes into creating good problems, assignments, and labs. Creating challenging and rich study material in computer science is different from any other subject, let's say, math. And it's not to say that math teachers don't spend a lot of time crafting problems as well. I've seen Mr. Kats and Mr. Cocoros sit down and really work things out for a whole period to create a good problem. I'm choosing it because I taught it for a while. Say you learn about the Pythagorean Theorem. It is helpful to practice the theorem a bunch of times. However, in computer science, if I want students to write a function to find the hypotenuse of a right triangle, it isn't really meaningful for me to have them repeat the task. It would just result in them writing the same code. But writing the same thing multiple times in CS has a high diminishing marginal return, and I generally have a rule that I never have my students write the same thing more than once. Therefore, coming up with problems in computer science is difficult because it requires creating bigger frameworks so that students have both room to explore and practice.

This concept has come up a lot for me recently with the NeXTCS class, which is a new course. I had my students create sorting algorithms on Processing that visually moved blocks around based on their sizes. The purpose of the assignment was for students to learn and write the sorting algorithms, and I had to prepare all of the visual stuff to ensure it would work. As long as the students wrote their sorting algorithm correctly, they could focus on the thing that they were supposed to be learning. Though that's just a particular example; this shows up in CS classes all over. Mr. Holmes is famous for spending lots and lots of time producing homework because he wants it to be the right thing for students to work on. The amount of work that goes into producing good assignments is great, even if they're small assignments. So maybe that is the most unappreciated piece.

3) Could you ever see yourself teaching another subject or working in a completely different job?

Teaching a different subject? No. My teaching career started 17 and a half years ago as a math teacher because that was the only avenue available to me. Then, 15 and a half years ago, I came to Stuy, switched to teaching computer science full time, and never looked back.

There are plenty of subjects that I would enjoy learning. I was a philosophy major in college, so I'm not a stranger to the humanities. But in terms of teaching, I love teaching computer science. I couldn't see myself teaching anything else. Even if I teach the same classes ten years in a row, such as graphics, there are always new programming challenges that keep the teaching interesting and enjoyable. It's not like I'm just opening up my old lessons and teaching the same things.

In terms of seeing myself do something else for a living, probably not. I do enjoy programming, so maybe there's a world where I could do some software development. Nonetheless, despite all of the problems that exist in the teaching profession, I enjoy it far too much to leave it in this lifetime. I'm not just writing code; I'm interacting with students. If you're a teacher and you're doing it for long enough, you're sticking around usually because there's something about being in a room with 30 students and working with them that is rewarding. It's not the vacations, it's not any of the silly things. It's working with students.

4) What's one advice you would give to students? It doesn't have to be specific to computer science.

Well, I think a piece of computer science advice that is also a good piece of life advice is that there is very rarely a "best" anything. Students will ask, "Oh, what's the best programming language? What's the best kind of computer? What's the best code editor?" The answer is that nothing is the best. Certainly, some are better for specific tasks, but there are far too many of any of those things for any one of them to be supremely the best. There was a period of time when I was younger when I would fall into this trap of being like, "Oh, well, this is clearly a better one." But in reality, you should recognize the good qualities in any of these tools, or whatever they might be. And also recognize that just because someone prefers one over the other does not define their own worth. Just because someone really likes programming in Java doesn't make them any better or worse than someone who really likes to program in Python. Saying this in the context of computer science is important, but it extends to all of life in general. Rather than focusing on what is best, we should concentrate on learning about a wide range of experiences and understanding.

The *Star Wars* phrase "Only a Sith deals in absolutes" comes to mind. The world is complex. There are good and bad things about all sorts of things. We should be able to recognize that and find it worth having, regardless of your own personal feelings. No matter their preferred subject—musical tastes, movie tastes, or any of that—everyone has value we should appreciate.

5) Through your own experiences teaching at Stuy for over 15 and half years, how have you seen the school change?

The 2007-2008 school year was my first year teaching here, and shortly before that time I was a student here, graduating in 2001. So I've been here for a total of nearly 20 years. One of the positive ways that Stuy has changed is that students have become more accepting of different experiences and different types of people. I think that there is more of a sense of people being able to find their place. Stuyvesant has always had more than 80,000 clubs and 40 sports teams, providing a whole range of different communities where students can find their interest and expand it. Stuy Squad is something that comes to mind; that didn't exist when I started teaching here. It's this thing everybody enjoys doing, and I don't know the real evolution of it, but I imagine people really enjoyed doing the dance stuff in SING! and made it a whole separate event.

However, during that same period, the level of competitiveness and focus on grades increased to the detriment of learning. Though it has always existed, it seems harder for students not to get sucked into the competitive atmosphere. Everybody thinks that they've got to be better than the other students in order to get ahead in class, in college, in life and so on. I think that that has gotten somewhat worse over time.

6) How do you think your own teaching has changed over 17 years?

I've improved my ability to listen to students and determine the value in the way they see problems or potential solutions, even if they aren't the things I expect to be included. For example, let's say I have a lesson about binary search. I first want to have a discussion about it, and eventually get to talking about its actual algorithm. Earlier in my career, when students would have that type of open discussions, I was more likely to brush those off instead of giving their ideas a moment to be in the class and be heard. I've gotten better at respecting ideas that are not what I would have expected or where I wanted lessons to go.

I've really been trying to promote a judgment-free zone, more so as my career has progressed. I happen to very much enjoy *Star Wars* and *Star Trek*, and early on in my teaching career, if the student said, "Oh, I've never seen *Star Wars*," I would be like, "What do you mean you've never seen *Star Wars*?" The fact of the matter is, it doesn't matter whether or not you've seen *Star Wars*. It doesn't affect whether you'd make a good computer science student or teacher. Now when I teach, I make it clear that it is something that I enjoy, but it doesn't have to be this "I can't believe you've never seen this." I don't want those sorts of unofficial barriers, and I work to ensure I am not creating any myself.

7) How do you spend your time outside of teaching? What do you enjoy doing the most?

Well, what I spend the most time doing right now is spending time with my son. I have a three-year-old, and that is all the time that I have, for the most part, in terms of my free time. And it's wonderful, but it is hard and difficult. Readers of this publication are young, so you're probably not looking into having children anytime soon, but don't do it quickly. It's a lot of work, even though it is rewarding.

But if I'm not hanging out with my son, I'd like to spend my time with friends playing board games, like tabletop role-playing games. I have a group that I've been playing pen and paper role-playing games with for almost three years now. We played a *Star Wars* game for a couple of years, and now we're doing a superhero-style game, such as *Dungeons and Dragons*. And in a similar vein, there's a phenomenal *Star Wars* card game that I had a really good group of people to play with. There was a time when I was competitive nationally. It's not a massive game like *Magic*, *The Gathering*, or *Pokemon*, but it's something I spent a lot of time playing and became quite skilled at.

Questions and Answers with
Distinguished Guest Dr. Jeffrey N. Heinz:
The Role of Mathematical Linguistics, the Rise of A.I.,
and His Favorite Projects

Shaon Anwar



Dr. Jeffrey N. Heinz is a Professor at Stony Brook University with a joint appointment in the Department of Linguistics and the Institute of Advanced Computational Science. He conducts research in several related areas including theoretical, computational and mathematical linguistics, grammatical inference, computational learning theory, theoretical computer science, robotic planning and control, and artificial intelligence. His research has been published in the Journal of Language Modelling, Linguistic Inquiry, Theoretical Computer Science, the Transactions of the ACL, and Science. He has co-authored a book on grammatical inference for computational linguists, co-edited three books, and co-guest-edited special issues of the journals Machine Learning and Phonology.

Professor Heinz graduated in 1996 from University of Maryland with a B.S. in Mathematics and a B.A. in Linguistics. He obtained his Ph.D. in Linguistics from UCLA in 2007 and spent ten years as a professor at the University of Delaware before coming to Stony Brook in 2017. The Linguistic Society of America recognized Professor Heinz with its 2017 Early Career Award for his “Contributions leading to a new computational science of inference and learning as applied to language.” In 2022, he received SUNY Chancellor’s Award for Excellence in Scholarship and Creative Activities. When not busy teaching and researching, Professor Heinz enjoys swimming, playing chess, solving puzzles, and programming.

Professor Heinz’s answers have been taken from an interview conducted by Shaon Anwar and have been edited for length and clarity.

1) Linguistics is an inherently interdisciplinary field. With its many subfields and connecting disciplines to focus on, you are investigating the sound structure of language from a mathematical and computational perspective. What drew you to this area of research?

When I first took a linguistics course as an undergraduate, I felt phonology was, in some ways, a bit more concrete than other areas like syntax. It was partly because I wanted to be a mathematician, so I came into linguistics for mathematics. I was kind of skeptical about what professors were saying about syntax and tree structures, whereas in phonology, it just seemed a lot more clear.

The other part of it was that in phonology, a lot of it is completely unconscious. When I'm talking, my stress patterns are a certain way, everything is a certain way, but no one taught me that. I read a lot of Bertrand Russell when I was your age. He wondered how it is that humans, whose encounters with the world are so brief and so fragile, come to know as much as they know. That was really striking to me. One of the things we know is our language, which is actually quite an intricate set of knowledge that we have, and again, a lot of it is unconscious. So it seemed to me that trying to understand how we come to acquire that phonological knowledge was a concrete way we could begin to ask some parts of the question that Russell was asking.

2) Did studying mathematics as an undergraduate student prepare you in the field of linguistics?

I think it helped me prepare a lot! I came in as an undergraduate only wanting to study math. I wanted to be a pure mathematician, and like I mentioned before, I was interested in Russell, so I was interested in Proof Theory. I felt like at the time, I hit a wall in mathematics. I was in an advanced class, and I didn't do well in it. I withdrew from the class, and I felt like my dream of being a mathematician was over. Then, I took a linguistics course, and I thought it was an easy course, but I could also see that there was a system there. Mathematics is really just a way of thinking about things. Obviously there's knowledge there, but it's really a kind of attitude more than anything: how you approach something.

I will also add that I had the pleasure of working with a lot of students who have done a lot of work in mathematics and went into linguistics. I find that, if they're interested in languages, the linguistics courses offer no resistance. They can understand it so quickly. Linguistics, when compared to mathematics or physics, is a relatively young field, so you can really get to the cutting edge with just a couple of years of study, and not like in physics or mathematics where you really have to be an advanced student in high school, and advanced student in college, get a Ph.D., and do several post-docs.

3) How would you describe mathematical linguistics?

Linguistics is often described as the scientific study of language, and so we should think of language really as a natural phenomenon. In some ways it's a biological phenomenon because we are biological creatures, and we have the capacity to communicate at a very high level. We're talking about abstract ideas where even the notion of a word is an abstract concept. So, there's something interesting going on here about languages.

Language is also systematic; it's not just the case that any sequence of words is going to form

a sentence in English. So what is that system? How do we explain why some sentences can be interpreted one way, but not some other way? These are the kinds of questions that linguistics are interested in. What is the underlying system that governs language? Mathematical linguistics brings the mathematical way of thinking to these questions.

For example, we want to build a translator that expresses sentences in one language into another language. Ultimately, a language is a sequence of words or symbols. Think about when you study mathematical functions. First, you learn about linear functions, polynomial functions, or quadratic functions. Then you learn about trigonometric functions. By the time you get to calculus and integrations, you have a very rich vocabulary to talk about numerical functions. Now, consider functions that take strings, sequences of symbols, and convert them into other sequences of symbols, which is fundamentally what translation is like. What kind of function is that? It's not a numerical function; it's a function over sequences. Unlike the study of numerical functions, the study of functions of sequences is less than one hundred years old. Now, I would say we have little understanding of these functions.

It's almost like saying, "Well, we've got linear functions and nonlinear functions." It's a bit more than that, but it's an approximation. We're actually getting a much more fine-grain understanding of the types of string-to-string functions that there are. When we have that, we can say things like, "When we're translating from this language to this language, it's this kind of function," which is important! How does gravity work? Well, it's an inverse square law. So we need to know what that means. Right now, we're coming up with the vocabulary for the mathematics behind words that would mean things kind of like "inverse square law," but for functions over sequences instead of functions over numbers. That's one example of what mathematical linguistics is about.

4) Earlier you mentioned translation software. Recently, artificial intelligence breakthroughs including the AlphaFold software and ChatGPT have made waves in the scientific world. How will this flurry of innovations create solutions to problems in linguistics?

We have to understand what a solution really is. What ChatGPT does is obviously very impressive, but we can ask, "What problem is it solving?" And that's a harder question to answer. Again, ChatGPT has been trained on incredible amounts of data. That's not what human children are doing when they learn. You weren't exposed to every book that's in the Google Book's library that's written in English. You learned to speak and generate and understand sentences that you never heard before within the first few years of your life. Your linguistics experience was sufficient for you to get a fairly good command of the English grammar. That's not how ChatGPT works. ChatGPT is doing a lot of statistical inference, and I'm not saying statistics is the wrong answer; I think statistics is going to be part of the answer. The problem with statistical methods is that rare events are difficult to deal with, and when you're talking about the kinds of sentences that we can understand and produce, there are a lot of rare events.

For example, a Zipfian Curve is an inverse power law, so the frequent things are very common, but the infrequent things actually dominate. A common example has to do with the frequency of words in texts. If you take any English text and count the number of times each word occurs in the text, you'll find the word "the" is by far the most common English word. Then, if you go down to the second word "of," it occurs half as much. The point is that as you keep doing this, you'll find that most words in the text occur exactly once. Most constructions occur just once, and so

how do you distinguish between the things that are rare and just occur once, and the things that are impossible? What these systems are good at is the first part of the inverse power law; they're not good at the tail.

Now, have they made progress? Yes, I would say there's definitely been progress, but the idea that somehow there's nothing else to do is not true at all. The other side is: are they fundamentally doing the right kind of work that's going to lead us to a genuine solution? That, I think the jury is out on. I think it's easy to say that they have to take this further than anything else, but look at the teams that are behind these teams. If I had those same resources that those teams have, using other kinds of methods, I think I can get pretty far too! We can't compete with them; they've got incredible resources behind them, and they're pursuing a particular line of research. They're getting somewhere - that's great. I just think that there's another way to go that will help us get real solutions to some of these problems. It's fine to disagree.

5) What are some of your favorite projects?

I loved working on my dissertation because I was answering a question about phonological patterns and how they could be learned. Since then, I've worked on projects about kinds of string-to-string functions; something that will have a very lasting impact. We're in the process of figuring out the fundamentals of it now. Again, one of the exciting things about mathematics is that you get to figure this stuff out, and if you get it right, it's there. Ken Dill, who is one of the faculty members at Stony Brook University, had come up with a very simple way of posing a protein folding project that I worked on, which is a very simple puzzle. You consider a sequence of red and blue dots - beads on a string. That's your sequence of amino acids. Basically your amino acids come in two basic types, the polar ones and the hydrophobic ones. One of the main factors in protein folding is that the hydrophobic amino acids - the blue ones on your string - want to clump together. So now you have a sequence of blue and red dots and you want to fold them up to maximize the number of blue-on-blue contacts. His proposal was to put them in a 2D gridspace, and now you want to put this string somehow along the grid so that the number of blue dots that are adjacent is maximized. That's actually a really hard problem, but you can use computational linguistics methods for parsing. What do we do in computational linguistics? We have a sequence, and we try to get a structure out of it because the structure tells us something about how we interpret it and how we understand it. That's what we're doing in this protein folding problem; we have a sequence of amino acids and we want to get a structure out of it: we want to fold it up into a 3D object. The structures are different in linguistics and protein folding, but the fundamental problem of taking a sequence and getting a structure out of it is very, very similar. They use the same kind of methods, these parsing methods, and if you can find a way to efficiently parse these amino acids into structures, it's going to be huge.

One of the great things about working on mathematical linguistics is that mathematics, because you've abstracted a way to a certain degree, allows me to work on these interdisciplinary topics, and it allows me to communicate with the protein folding guys, and the robotics and planning guys. It's been fascinating to me. I've learned so much, and I love it! It's one of the great things about the institute I'm in at Stony Brook; it's an interdisciplinary institute, and that's been really fabulous.

6) What are the challenges you encounter in your field of research?

I mentioned before that in linguistics there are different communities of people who have different opinions about things. Frankly, a lot of people don't think the mathematical aspect is very useful. For them, it's too abstract. They think it doesn't connect with reality. What I would say to them is that everything is abstract! A lot of them would prefer to use statistics or real numbers or things like that. But, numbers are abstract. Real numbers are especially abstract. The idea that somehow what we're doing is too abstract, I don't really understand.

I will say that within and beyond the linguistics field, people want to see applied research. There's nothing wrong with applied research - doing applied research is great, a lot of technological advancement we have in the last 50 years is coming from applied research, but I don't think applied research is the only kind of research that should be done. I think theoretical research should also be done, where the horizon is a bit further down the road. The work that I'm doing now, these fine-grained understandings of functions over sequences is going to be integrated into all the technology not in five years, not in ten years, it's going to be like 30 years from now. It's going to take a while for it to get established. It's going to take a while for people to realize it's out there, then they will start using it, then it will get integrated. So, some of the challenges I face in my own research is that people are skeptical of the mathematical stuff. My own view is that there are many ways to search for the truth, and people should welcome different approaches even if they are more theoretical than what they are comfortable with.

7) Lastly, what is your advice to Stuyvesant students who would like to pursue computational and mathematical linguistics?

You're so lucky! It's a great time to do that. The mathematical and computational stuff is more accessible than it has ever been before. Keep an open mind, study the questions that interest you, learn as much math and computer science as you can, but also learn as much about languages as much as you can! I had a professor, Ed Keenan, who was a mathematical linguist at UCLA, and he would say, "You should read grammar like you would read science fiction." You should go into the library; there are stacks and stacks of shelves and there are whole shelves that are filled with grammar. You start looking at it and ask yourself: "What does this language do? How does it do this thing? What does it do here?" It's almost like a botanist studying the different kinds of things plants do. You want to get a sense of that, because then all the mathematical and computational concepts will begin to sort through them in your mind. I would say that it's a great field to go into.

Google Panel 2022: Peering into the Work World of Computer Scientists

Anastasia Lee



On December 13th, were you in Murray Kahn Theater to learn about some of the inner workings of the Google offices just a subway ride away on 8th Avenue? Thanks to Mr. Dillon, computer science teacher in his first year at Stuy, and Ms. Prabhu, Assistant Principal of Biology, Computer Science and Special Education, a panel of four guest speakers gave Stuy students and faculty members an insightful glimpse into some different roads and different roles within the tech giant. Tamiyyah Shafiq and Alexander Panas, sophomores from Mr. Dillon’s CS class, served as welcoming hosts for this event by providing opening remarks, introducing the panelists, and coordinating the Q&A.

1 Why Host a Google Panel, and Why Now?

In an email interview, Mr. Dillon described his motivation for initiating this opportunity, saying, “Computer Science Education Week (CSEdWeek) is an annual event during which teachers engage students to learn computer science. Schools around the country host special events like hack-a-thons, Hour of Code, or guest talks with people in tech. Industry partners often get involved, which is where Google came in. My motivation... was to share the guest speakers’ stories about their path to a career in tech. I’ve always been a teacher, so I don’t have a personal story about finding a job in a tech company like Google.” He credited Ms. Prabhu for her crucial help with a wide variety of logistics, as well as a Google Program Manager, Mr. Dion Paul, for serving as liaison and providing the panel members who generously volunteered their time and their stories.

So who were these informative Googlers who shared their backgrounds and their industry perspectives with us? **Mr. Andrew Myers**, a network architect with Google Distributed Cloud

Hosted, is a “Noogler,” having worked for Google for nine months, designing networking systems that can quickly transfer data reliably and efficiently. He earned an M.S. in Applied Mathematics from Odessa University and an M.S. in Management from Harvard University. **Ms. Catherine Liang**, a software engineer with the Android System User Interface team, also a “Noogler,” works with designers to implement design and functionality changes to the Android OS. She graduated last May from the University of Pennsylvania with a B.S. in Computer Science and Computer Graphics. **Mr. Benjamin Taubenblatt**, has worked with Google for two years as a software engineer on its Speech Recognition Infrastructure. He builds systems that support speech-to-text technology for products like the Google Assistant. He graduated from McGill University with a B.S. in Computer Science and Linguistics in 2019. **Dr. Pratik Worah**, a senior research scientist with Google Research, has worked at Google for about eight years on probability theory, neural networks, and artificial intelligence underlying algorithms. He earned a Ph.D. in Theoretical Computer Science from the University of Chicago in 2013, following his undergraduate studies in Computer Science and Engineering at Indian Institute of Technology, Kharagpur and his M.S. in Applied Mathematics at University of Illinois, Urbana.

2 Googlers and Their Stories

Each Google panelist took the mic to offer their individual career paths, starting with Ben Taubenblatt. Surprisingly, he initially had a negative experience with math in high school, nearly dropping AP Calculus in senior year. “I was never someone who was super good at math, never studied the right way for it,” he recalled. But he changed how he approached the material and slowly improved, ultimately earning that AP credit, which inspired him to take higher level math and computer science classes in college and then to major in computer science. During college, he completed software engineering internships at FINRA and Morgan Stanley. After graduation, he worked in consulting at Deloitte, in its Big Data and AI department. After one year, at the start of the pandemic, he was assigned to business intelligence projects, but quickly realized that he was passionate about software engineering and programming, so he quit after just two weeks. Eventually, a friend referred him to job opportunities at Google, and he embarked on the lengthy application process that entailed hard programming/brain teaser interview problems.

Like many teams at Google, he codes primarily in C++. He builds systems for speech researchers at Google who host machine learning models, the core technology behind the conversion of a raw audio query into text. Specifically, his team handles the ingestion of large amounts of speech data for those models, the transcription and annotation of that data, and checking in data sets for training the models.

Next, Catherine Liang described how she began her “great experience” at Google. In college, she studied digital media design, including e-design and automation. As a high school student in New Jersey, she was not exposed to much in the way of computer science education and was more interested in art. To be practical, she decided to pursue college studies in data graphics to open up tech job prospects while still utilizing her skills in art. She started as a Deloitte tech consultant, working on infrastructure and setting up JavaScript frameworks.

When a Google recruiter reached out to her after her first year at Deloitte, she was already interested in applying, since she had always wanted to work for Google in college. She went through three rounds of technical interviews and one round of behavioral before she was matched with the

Android OS UI team. Using Android Studio, Java, and Kotlin, she contributes to various aspects of the frontend user experience. To share a fun fact with us, she mentioned that she created games using Unity and Unreal Engine!

Andrew Myers came prepared to show off his brightly-colored Noogler hat, complete with a small plastic propeller on the top! He is still learning about Google culture in his first year by asking a lot of questions. Despite his significant experience in systems at large companies such as Morgan Stanley, Microsoft, and AT&T, his journey to Google employment was “not quick,” receiving an offer only after persistently applying three separate times (five years ago, two years ago, and less than one year ago). He emphasized that foundations in both math and computer science are equally important to be successful in tech.

In his team’s work designing networks, he is building the foundation of Google Cloud, and he uses the platform Anthos extensively. One important aspect of network design is security, including protection from both internal and external attacks, as well as compliance with government requirements. He offered this accomplishment as a fun fact about him: He once rebooted just one router to restore Internet service to the whole tri-state area!

Pratik Worah shared that he grew up in India and was originally “crazy about physics.” He admitted that when he first studied computer science, he was “right at the middle” of his class, while his classmates were much more familiar with programming in C. But he continued with his math and computer science studies on “more or less autopilot” until he completed his Ph.D. in Theoretical Computer Science. Much of his research focused on complexity theory, which involves proving impossibility, one of the deepest areas of mathematics.

After a little over one year of post-doc work at NYU’s Courant Institute of Mathematical Sciences, he spent slightly over one year at a global finance firm, PIMCO (Pacific Investment Management Company), where his research involved mathematical models for bond and derivative investment strategies. He has enjoyed working on a diverse portfolio of problems, some involving economics or game theory, and commented on the applicability of calculus to find maxima and minima. Although he had worked briefly for Google in Mountain View as a software engineer in between getting his Master’s degree and his Ph.D., it was not until his colleague encouraged him to work for Google Research that he left his job as a quantitative researcher at PIMCO and began his work as a Google Senior Research Scientist, focusing on neural networks and their limitations.

3 We Had Questions, They Had Answers

Attendees kept the discussion going with their questions for the panelists, submitted via QR code. When asked for advice about how to get **internships**, Ben Taubenblatt remarked that internships provided him with his first taste of real, “out in the wild” programming at a company, where he saw how teams and companies operate. He acknowledged that getting the first internship is the hardest, since you don’t have any work experience on your resume. He encouraged students to make their resumes stand out by doing personal projects, which show passion. For example, if you can build a simple webpage, both the front end and the back end, and include a URL on your resume that a recruiter can click on. He advised that your resume should be concise, one page, since recruiters at large companies only spend 10 to 20 seconds on each one, and you want the most important items to stand out.

Catherine Liang reflected that she had a different experience than many of her peers, since she didn't intern at any big tech companies, like most software engineers. She advised that it's good to start interning early, noting that some of her classmates interned during the summer after freshman or sophomore year, and were able to get return offers for the summer after junior year, leading to a full-time offer after graduation. She also encouraged students to try their best when they apply for internships, even though the process is highly competitive. But she noted that if an internship opportunity doesn't work out, she hopes students "get more excited about doing your own projects." For example, she gained software development experience by working on fun game design with friends over the summer.

Andrew Myers emphasized the importance of starting to build a network "from the beginning," by getting to know people in tech companies. That way, when the time comes to get internships, your network can help you get a head start over others.

Students also inquired about the most **enjoyable/rewarding parts** of working in the panelists' different fields. Andrew Myers noted how network engineers can bring back lost foundational services and allow vital computer systems to withstand drastic events. Catherine Liang remarked that her software engineering work on frontend Android functionalities makes smartphone use "bug free" and more easy and enjoyable. Ben Taubenblatt shared that he loves how the coding process in general is "extremely creative" and that problem-solving does not just involve math and logic. He described how people approach the same high-level problems completely differently. He finds his work of developing solutions to support the machine learning models of speech recognition systems does not have a "defined playbook," so Google is a trailblazer on the leading edge of that technology.

Lastly, audience members asked panelists what they look forward to the most in the **future of computer science**. Pratik Worah predicted that computer science will become even more foundationally useful in more diverse ways but likened computer science to a screwdriver for a mechanic. He opined that computer science doesn't have to be the only focus of life, saying, "No one studies a screwdriver!"

Ben Taubenblatt looks forward to the future of quantum computing, "the next big jump." Just as transistors allow the smartphone in our pockets to be more powerful than the computers from the 1950s which took up entire rooms, he predicts that having the power of quantum computers in our pockets will be incredible.

4 Our Attitude of Gratitude

The Stuy CS community sincerely thanks these Googlers for their enlightening and enthusiastic information and advice. Mr. Dillon reflected on the panel discussion, saying, "I think the guest speakers did a great job of sharing stories of perseverance and hard work with our students. Many of my students who attended found those stories inspiring and took time after the event to meet the panelists."

To learn more about Computer Science Education Week and its "call to action," we encourage our readers to visit <https://www.csedweek.org/> and explore all the various resources and opportunities.

The Year in Math

Distinguished Guest Contributor Konstantin Kakaes,
Senior Editor in Charge of Mathematics Coverage at Quanta Magazine

Four Fields Medals were awarded for major breakthroughs in geometry, combinatorics, statistical physics and number theory, even as mathematicians continued to wrestle with how computers are changing the discipline.

This article was reprinted with permission from Mr. Thomas Lin, Editor-in-Chief of Quanta Magazine, an editorially independent publication of the Simons Foundation. This article can be accessed via: <https://www.quantamagazine.org/the-biggest-math-breakthroughs-in-2022-20221222/>.

We can think of a mathematician as a kind of archaeologist, painstakingly brushing dust off the hidden structures of the world. But the structures mathematicians reveal are not only durable, but also inevitable. They could never have been any other way. They are also remarkably interconnected: Though each year the mathematical frontier continues to expand as new discoveries are made, the sprawl of subdisciplines also shrinks a little as connections are found between seemingly far-flung domains.

It can be difficult to convey to the nonexpert just how shocking some of these connections are. In 2022, as always, we've done our best.

It was a year of short papers — a six-page proof that pinpointed when structure emerges in random graphs — and long ones — a 912-page paper showing that slowly rotating black holes are going to keep on slowly rotating until the end of time.

Some mathematical results can be impenetrable not only to the public but also to other mathematicians. Quanta spoke to Lillian Pierce, a number theorist at Duke University, about her work to make important proofs and techniques understandable to a broader range of mathematicians, and to Wei Ho, who discovered new bounds on the number of integer solutions to equations called elliptic curves.

Alex Kontorovich, in a video and accompanying column, discussed the sweeping Langlands program that has been making connections across disparate fields of mathematics.

Though many of these results have no immediate practical applications, it's tough to know what abstract result will end up becoming crucial to coming up with new, secure cryptographic ciphers or updating the error-correcting codes needed to make modern communications work.

One thing that is clear from the past year of math coverage in Quanta is that there is no one path to becoming the type of mathematician who can uncover a fundamental truth that nobody else could. Some are singularly focused on mathematics from a young age; others come to it later

in life. Some satisfy a stereotype of absent-minded brilliance; others don't.

As Huh put it, when it comes to how the human mind makes leaps of mathematical reasoning, “it’s nice to admit that we don’t know what’s going on.”



June Huh with a polyhedron.
Caroline Gutman for Quanta Magazine

1 Award-Winning Mathematics

Every four years, the International Mathematical Union awards a gold coin with the head of Archimedes engraved on it to up to four mathematicians younger than 40 “to recognize outstanding mathematical achievement for existing work and for the promise of future achievement.” This year the Fields Medal went to June Huh, James Maynard, Maryna Viazovska and Hugo Duminil-Copin.

In a profile, Huh explained that “mathematics could give him what poetry could not: the ability to search for beauty outside himself, to try to grasp something external, objective and true.” The prize recognized his proof of a number of different conjectures. Among other things, he found a deep geometric structure “lurking beneath combinatorial properties of graphs” when he proved something known as Read’s conjecture.

Maynard’s award was for his discoveries in analytic number theory. Not long after he finished graduate school, he proved that there are an infinite number of prime pairs that differ by 600 or less. This was a landmark result, but it would have been even more significant if another mathematician had not proved the existence of a finite bound on the gaps between prime pairs a few months earlier. It was a remarkable example of parallel progress on a problem related to questions about the distribution of prime numbers. But Maynard added to his work on prime gaps by proving that there are an infinite number of primes that don’t contain a given digit, like 7.

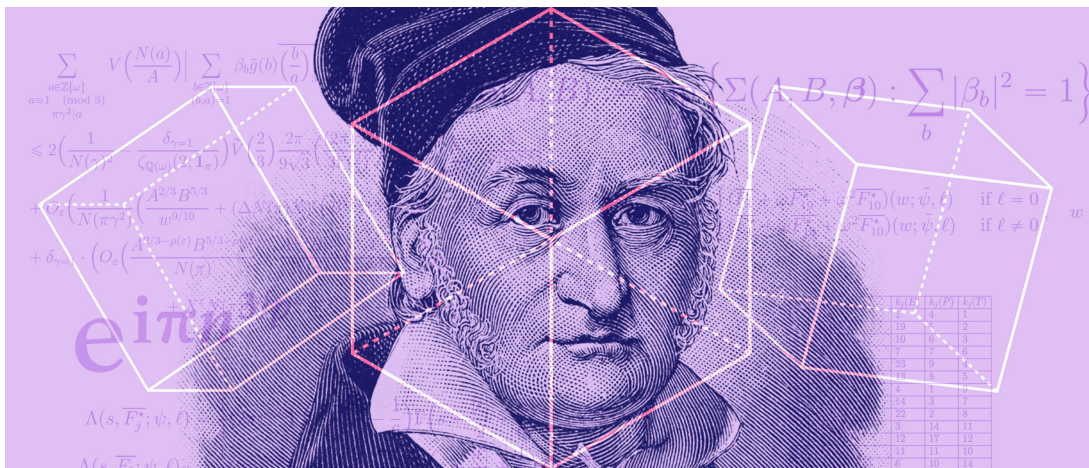
It’s long been known that the densest way to pack circles in a plane is in a honeycomb, and a decade ago a conjecture that had stood since the 17th century about how to most efficiently pack spheres in three dimensions was proved. But higher dimensions are mostly a mystery. Viazovska

proved that a particular eight-dimensional lattice provides the most efficient way to pack spheres in eight dimensions. She and collaborators then generalized that result to prove that this lattice minimizes the energy of a system in a wide variety of contexts.

Duminil-Copin, the fourth medalist, won for coming up with a generalized theory of how liquids flow through porous media.

The Fields Medals weren't the only prizes awarded this year. Dennis Sullivan won the Abel Prize for his contributions to topology, which included coming up with a new way to classify certain types of manifolds — spaces that look flat on a small scale but are more complicated when examined in their entirety.

And for achieving a better understanding of so-called quasi-periodic operators, which are used to model electron behavior, Svetlana Jitomirskaya was awarded the first Ladyzhenskaya Prize in Mathematical Physics.



Kristina Armitage for Quanta Magazine

2 New Proofs of Old Number Theory Problems

It was a bumper year for number theorists of all ages, following a productive 2021. A high school student, Daniel Larsen, found a bound on the gaps between pseudoprimes called Carmichael numbers, like 561, which resemble primes in a certain mathematical sense but can be factored (in this case $561 = 3 \times 11 \times 17$).

Jared Lichtman, a graduate student at the University of Oxford, showed that actual primes are, according to a certain measure, the largest example of something called a primitive set.

Two mathematicians at the California Institute of Technology proved a 1978 conjecture predicting that cubic Gauss sums, which sum numbers of the form $e^{\frac{2i\pi n^3}{p}}$ for some prime number p , always add up to about $p^{5/6}$. Their proof assumed the truth of something called the generalized Riemann hypothesis, which mathematicians widely believe to be true but have not yet proved. Meanwhile a

simpler analogue of the Riemann hypothesis called the subconvexity problem was solved.

A pair of mathematicians showed that whether an integer has an even or odd number of prime factors has no effect on whether the integer before or after it has an even or odd number.

Another group showed that at least $2/21$ and no more than $5/6$ of integers can be written as the sum of two cubed fractions.

In 1993, a mathematician named Peter Stevenhagen conjectured that the equation $x^2 - dy^2 = -1$ has an integer solution about 58% of the time under certain constraints on d . This year, his hypothesis was proved.

It was one of several long-standing conjectures that were shown to be true. The 30-year-old André-Oort conjecture about the structure of something called Shimura varieties was finally proved, as was the 85-year-old Van der Waerden conjecture, which estimates how many polynomials have noninterchangeable roots.

In the 1970s Paul Erdős and Ronald Graham posited that sufficiently large sets of integers must contain subsets whose reciprocals sum to 1, which was proved this year. It was also shown that such large sets of integers must contain something called an infinite sumset; this was proved using methods from the study of dynamical systems.

3 Machines Learning Math

Deep learning, the widely used AI technique that has beaten champions at games like chess and Go and proved extremely accurate at tasks like speech recognition, is also being deployed in some areas of math. Researchers used it to search for unusual singularities, points of breakdown in equations that model the flow of fluids. A separate team used computer-assisted proofs to definitively prove that a particular version of the Euler equations, which model certain types of ideal fluids, breaks down. Yet another group poked at the related Navier-Stokes equations, which more accurately model most real-world fluids, to see if they too might fail. (Anyone who shows that they do stands to win a million-dollar prize from the Clay Mathematics Institute.)

Several other groups used machine learning to solve problems in graph theory and combinatorics, create better techniques for multiplying matrices, and come up with new conjectures in knot theory. Sébastien Bubeck and Mark Sellke turned the tables by using mathematical techniques for analyzing neural networks to prove how big they must be to work robustly.

In episodes of *Quanta's Joy of Why* podcast, Steve Strogatz asked Kevin Buzzard if computers can be mathematicians and spoke with Melanie Matchett Wood about what it takes for mathematicians to truly believe a result has been proved.



Nadzeya Makeyeva for Quanta Magazine

4 Bubbles, Shapes and Spaces

Not to be left behind, geometers also had a busy year. In May, Emanuel Milman and Joe Neenan found out the shape of clusters of bubbles that can most efficiently enclose three or four volumes — in any number of dimensions.

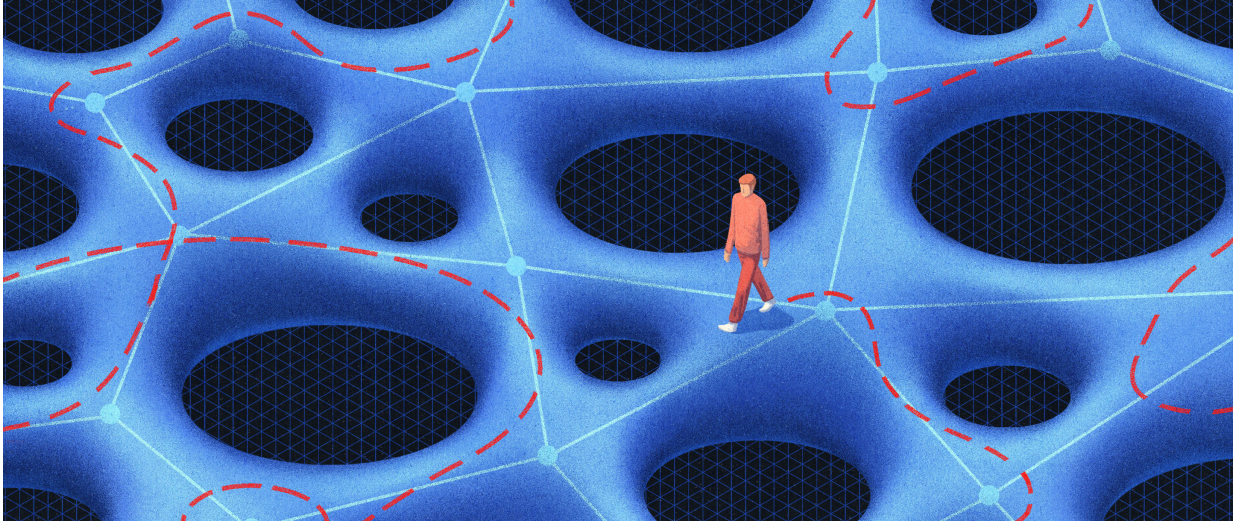
Isabel Vogt and Eric Larson solved the interpolation problem, a long-standing question about how many random points in high-dimensional space certain types of curves can pass through.

Andras Máthé, Oleg Pikhurko and Jonathan Noel solved an even older problem, figuring out how a circle can be cut up into visualizable pieces that can be rearranged into a square. (A *Quantized Academy* column explains the simpler process of figuring out whether two polygons have equal area.)

Martin and Erik Demaine (father and son) published a paper that shows how to take any polyhedron and fold it into a flat shape — as long as you allow infinitely many creases. (A *Quantized Academy* column explained how tetrahedra — pyramids — have much more intricate relationships between their interior angles than do triangles, their two-dimensional analogue.) In August, a collaboration of mathematicians and physicists published a paper that laid out a new theory of how the curvature of thin materials affects the wrinkles that are formed when they are flattened.

Dusa McDuff and several collaborators found intricate fractal structures emerging when they tried to embed shapes called ellipsoids into something called Hirzebruch surfaces — a place where nobody had expected to find fractals.

Other mathematicians made progress toward proving the Kakeya conjecture, which puts limits on how big a space is needed to turn a needle around so that it can point in any direction — but there are doubts as to whether the conjecture is in fact true in the domain of real numbers.



An illustration of a person walking across a surface with many holes.

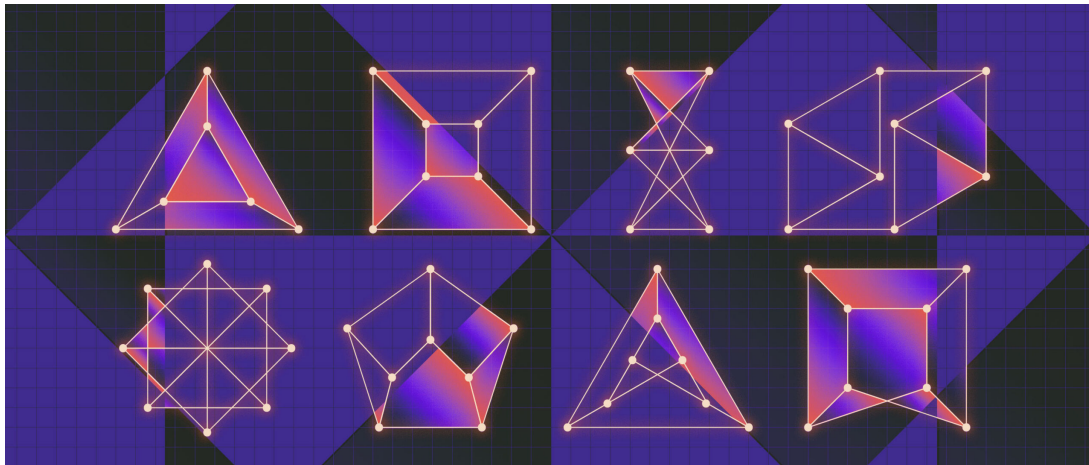
Myriam Wares for Quanta Magazine

5 Adventures in Topology

This year, *Quanta* covered the work of Will Hide and Michael Magee, who in 2021 used techniques borrowed from graph theory to show the existence of types of high-genus surfaces that are deeply interconnected with themselves in a way that had long been thought, but not proved, to be possible.

Ian Agol proved a 1981 conjecture about how to rank the complexity of knots.

Knots in two dimensions can be used to delineate the boundaries of something called a Seifert surface. Many pairs of Seifert surfaces that are distinct from each other even if the knots are manipulated in three-dimensional space can be made equivalent if the knots are manipulated in a fourth dimension. Topologists found, for the first time, a pair of Seifert surfaces that remained distinct from each other even in four dimensions. A column explained knot theory's origins, and Colin Adams and Lisa Piccirillo told Steven Strogatz about their fascination with knots on an episode of the *Joy of Why* podcast.



Points connected by lines.
Kristina Armitage for Quanta Magazine

6 The Emergence of Random Structure

A remarkably short proof published in March proved the Kahn-Kalai conjecture, which laid out conditions under which structure emerges in random graphs. This followed a proof in January that it's always possible to build a “hypergraph” — a sort of highly connected generalization of a graph — in a way that satisfies two seemingly incompatible criteria, as long as you make the hypergraph big enough.

New graph theory results kept coming. In April, Oliver Janzer and Benny Sudakov answered a half-century-old question about when graphs must inevitably become “regular,” or interconnected in such a way that every node is connected to the same number of edges. And a column gave an introduction to graph theory through an ice-breaking party game.

